

Be Green Hero(BGH) Token White Paper

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A Knowledge-First Sustainability and Governance Ecosystem

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Executive Abstract

BE GREEN HEROES (BGH) is a global **Learning-to-Earn ecosystem** designed to advance science-based environmental sustainability education and convert verified knowledge into meaningful, real-world action. The BGH Program addresses a critical weakness in today's sustainability landscape: the dominance of misinformation, oversimplified narratives, and decisions driven by perception rather than rigorous lifecycle science.

BGH introduces **Green Hero Coin (GHC)** as an incentive mechanism that rewards participants for completing structured learning modules, passing assessments, and engaging in mission-aligned activities. Unlike conventional incentive systems that amplify ideology, speculation, or popularity, BGH rewards **verified understanding and sustained educational contribution**. The model includes provisional participation mechanisms that recognize learning achievements regardless of age, ensuring that access to education and civic participation is not constrained by market eligibility.

The BGH Program is designed to evolve through a **staged decentralization framework**, in which governance authority is progressively transferred from the BGH Organization to a community of Green Heroes (GHs) who earn governance-enabled Utility Coins (UCs) through verified learning. Governance participation is grounded in demonstrated knowledge rather than political influence, commercial pressure, or financial purchasing power. Provisional Utility Coins (p-UC), earned by underaged participants, further extend this model by enabling governance participation while remaining strictly restricted from trading, transfer, or other market-based execution until eligibility conditions are met. Together, these mechanisms establish a new governance architecture: **decentralized decision-making led by a science-informed public**.

To support global scalability without imposing disproportionate environmental costs, BGH operates on **energy-efficient digital and blockchain infrastructure** that avoids energy-intensive computation such as Proof-of-Work mining. The Program evaluates technology choices through a systems lens, prioritizing environmental benefit per unit of energy consumed. In this framework, the modest energy required to operate education, governance, and verification mechanisms is understood as a high-leverage investment; one that enables informed decision-making, reduces the long-term environmental harm of misinformation, and supports durable, science-aligned sustainability outcomes at global scale.

Overview

BE GREEN HEROES (BGH) is a global **Learning-to-Earn** ecosystem designed to advance science-based environmental sustainability education and convert verified knowledge into meaningful action.

At the core of the **BGH Program** is **Green Hero Coin (GHC)** - a tokenized incentive model that rewards verified learning rather than hype, ideology, or marketing narratives. Participants earn tokens by completing educational content and demonstrating scientific understanding through structured assessments and mission-aligned engagement, **including provisional participation mechanisms designed to responsibly recognize learning by underaged Green Heroes.**

Unlike traditional incentive systems, BGH is built to become a **fully decentralized program through a staged decentralization model**. Governance begins with structured safeguards to protect scientific quality, then progressively shifts to the public as Green Heroes earn Utility Coins (UCs) through verified learning. This enables a new governance architecture: decision-making driven by an informed community—rooted in science-based knowledge rather than centralized control, political agendas, or commercial influence.

* Refer to **Appendix A – Disclaimer** for details. This applies to all mentions of financial benefit or gain in this document.

** Refer to **Section 8.3 (Gradual Transfer of Decision-Making Power)** for additional detail.

The Problem

Today's global sustainability efforts are frequently shaped by strong narratives—but not always by strong science. Key challenges include:

1) Misinformation and Oversimplification

Many people believe they “already understand” sustainability, while their views are shaped by myths, slogans, and viral content rather than scientific evidence and lifecycle-based analysis.

2) Low Motivation to Acquire Accurate Knowledge

Building meaningful scientific literacy requires time and effort. Without a clear incentive, most individuals do not invest the work required to develop reliable, decision-grade sustainability knowledge.

3) Science Gaps in Environmental Decision-Making

Even major organizations and environmental initiatives may prioritize perception over data. This often leads to misaligned priorities, inefficient allocation of resources, and sustainability outcomes that fall short of real-world benefit.

The result is a sustainability discourse that is highly visible—but not consistently accurate, effective, or measurable.

Our Solution

The BE GREEN HEROES (BGH) Program aligns personal incentives with science-based learning by integrating **education, verification, low-energy digital infrastructure, and tokenized participation** into a single, coherent system. BGH is designed to ensure that the environmental cost of operating the platform remains proportionate to, and justified by, the environmental benefits it enables.

1) Learning Platform

BGH delivers curated, science-based environmental sustainability education through structured learning modules designed to build accurate, practical understanding rather than superficial awareness. Educational content is selected, reviewed, and evolved through governance mechanisms that prioritize scientific credibility and lifecycle thinking.

The Program launches with a practical starting focus on textile sustainability—an area of high global relevance and environmental impact—and is designed to expand over time into additional high-impact environmental domains.

To support scalability without unnecessary environmental burden, the BGH Platform operates on **energy-efficient digital and blockchain infrastructure**, ensuring that the delivery of education and governance does not rely on energy-intensive computation or wasteful system design.

2) Learn-to-Earn Mechanism

BGH incentivizes meaningful learning through a Learn-to-Earn model in which participants earn Green Hero Coin (GHC) through verified, mission-aligned activity rather than passive consumption or speculative behavior.

Participants may earn GHC by:

- completing structured learning modules and assessments
- demonstrating evidence-based understanding of environmental concepts
- participating in mission-aligned engagement and community activities
- earning **Provisional Utility Coins (p-UC)** for verified learning by underaged participants, subject to non-execution and non-commercial safeguards

All tokenized participation is supported by **low-energy infrastructure choices**, ensuring that incentive alignment and governance participation do not introduce disproportionate environmental costs relative to the educational and societal benefits produced.

3) Science as the Core Filter

Science functions as the foundational filter across all aspects of the BGH ecosystem. The Program is explicitly designed to strengthen:

- critical thinking and analytical reasoning
- evidence-based evaluation of environmental claims
- the ability to identify greenwashing, oversimplification, and misleading narratives

By combining rigorous scientific framing with decentralized governance and energy-efficient digital infrastructure, BGH addresses not only *what* people learn, but *how* sustainability knowledge is produced, validated, and scaled—reducing the long-term environmental harm caused by misinformation, ineffective action, and poorly informed decision-making.

Impact and Value Proposition

By connecting token incentives to validated scientific learning, **BGH** aims to:

- Upgrade public sustainability understanding, moving beyond slogans into real trade-offs, lifecycle impacts, and measurable outcomes
- Improve the quality of environmental decision-making across:
 - policy and regulation
 - corporate sustainability strategies and **ESG** initiatives

- consumer behavior and investment flows
- Build a global network of **Green Heroes** who influence:
 - public discourse and social norms
 - grassroots initiatives
 - market demand and capital allocation

Potential Financial Upside for Green Heroes

GHC is designed so that participants may benefit financially* as the ecosystem grows, utility expands, and market participation develops over time.

Token sustainability is supported through:

- long-term, governance-controlled market coin release
- mission-aligned Treasury actions
- operational discipline and transparency standards

* Refer to **Appendix A – Disclaimer** for more details.

Decentralized Operational Principles

BGH is designed to evolve into a **fully decentralized operational model** through a staged decentralization framework. **Governance** is executed through transparent **Proposal and Voting** protocols, with structured safeguards in the early phases to protect scientific integrity and platform stability. Over time, decision-making authority progressively shifts to the public as Green Heroes earn governance-enabled **Utility Coins (UCs)** through verified learning, including **Provisional Utility Coins (p-UC)**, which remain non-executable with respect to trading, transfer, or market-related actions until eligibility conditions are met.

Science-Based Public Governance

Eligible governance decisions are made through voting by those who hold governance-enabled **Utility Coins (UCs)**, earned through verified learning participation. Provisional Utility Coins (p-UC) earned by underaged Green Heroes confer governance participation rights, including Proposal and Voting eligibility, while remaining restricted from execution, trading, transfer, or market-related actions during their Non-Execution Period.

This structure is designed to progressively transfer operational control to the public—not based on influence, wealth, or popularity, but on validated scientific knowledge gained through the platform.

Governance and Market Separation

To preserve mission integrity:

- Utility Coins (UCs) hold governance eligibility (proposal and voting rights)
- Market Coins (MCs) are designed for trading and exchange only, with no governance rights

Immutable vs Changeable Scope

The **BGH Program** protects its core identity and mission by defining certain foundational elements as immutable. All other eligible operational areas remain open to improvement through decentralized governance.

* Core values, total issuance limits, and defined foundational mechanisms are immutable; other operational elements are subject to governance-based evolution (see Proposal and Voting protocols).

** Execution Period definitions and time-based frameworks are described in the relevant governance and tokenomics sections.

Roles and Structure of This White Paper

This White Paper defines the full structure of the BGH Program, including its mission, learning system, governance protocols, tokenomics, and treasury management.

This document covers:

- program philosophy and goals
- governance architecture enabling fully decentralized operations
- learning and engagement verification mechanisms
- tokenomics and utility model of GHC
- treasury design and long-term sustainability framework
- transparency and secure operations standards

Section Map

- Definitions and Terminology - **Section 1**
- Purposes and Goals - **Section 2**
- Learning and Engagement - **Section 3**
- Stakeholders - **Section 4**
- Official Public Launch and Immutability - **Section 5**
- Governance Framework - **Sections 6 to 8**
 - Governance principles
 - Proposal protocol
 - Voting protocol
- Tokenomics and Treasury - **Sections 9 and 10**
 - UC and MC structure
 - mining and redistribution
 - treasury protocol (TTP)
 - transparency and reporting
- Continuous Operations Protocol (COP) – **Section 11**
- Safeguard Protocol (SGP) – **Section 12**
- Governance Continuity and Inactive Participation Management (GCIPM) – **Section 13**
- Glossary and Appendices

Time Standard: All time references in this document follow Eastern Time (ET), Canada and USA (Standard time: UTC-05:00, Daylight saving time: UTC-04:00).

Reference - Abbreviation Code Conventions

The following conventions apply throughout this White Paper:

- **Nouns** are expressed using **uppercase abbreviations**, such as *Market Coin (MC)* and *Utility Coin (UC)*, with the sole exception of *Be Green Heroes (BGH)*, which is written in full capital letters.
- **Adjectival modifiers** are expressed in **lowercase**, connected to the noun abbreviation by a hyphen (“-”), forming compound abbreviations. Examples include *mined-Utility Coin (m-UC)* and *Market Coin-Original (MC-o)*.

These conventions are used to distinguish token classes, states, and attributes consistently across definitions, governance, and protocol descriptions. The full list of the abbreviated codes is provided in the **Glossary** at the end of this document.



Section 1. Definitions, Terminology and Marketing Assets

This Section provides definitions and terminology for the framework and core structural concepts of the **Be Green Heroes (“BGH”)** token program and ecosystem. Capitalized terms used in this White Paper shall have the meanings set forth below unless otherwise stated.

Certain definitions may appear in more than one section of this White Paper in order to ensure clarity and comprehensive understanding across all audiences.

1.1 Be Green Heroes (BGH)

1.1.1 BGH

“**Be Green Heroes**” or “**BGH**” represents the values, purposes, goals, and tasks described in this White Paper and serves as the primary identifier for the overall program as well as its related components, identifiers, subcategories, and operational concepts.

1.1.2 BGH Program

“**BGH Program**” refers to the overall program described in this White Paper, encompassing both:

- its abstract values and mission, and
- the execution of all associated activities, operations, and initiatives.

1.1.3 BGH Platform

“**BGH Platform**” refers to the platform and systems through which all related BGH operations are delivered, including educational content management, public engagement, community participation, and general program operations.

1.1.3.1 BGH Platform-online / BGH Platform-onchain

The portion of the BGH Platform that primarily relates to digital operations may be referred to as “**BGH Platform-online**” and/or “**BGH Platform-onchain**,” depending on context and circumstances.

1.1.3.2 BGH Platform-offline

The portion of the BGH Platform that primarily relates to real-world activities and operations may be referred to as “**BGH Platform-offline**.”

1.1.4 BGH Organization

“**BGH Organization**” refers to the organization responsible for managing and executing the BGH Program and operating the BGH Platform(s).

1.1.5 BGH Prefix Identifier Convention

Any future identifiers, designations, or terms required to define activities and operations under this program may use “**BGH**” as a prefix identifier, followed by distinguishable sub-tags consistent with the naming convention above.

1.2 Green Hero (GH)

1.2.1 Green Hero

“**Green Hero**” or “**GH**” refers to a person who engages in the BGH Program for the purpose of learning **science-based knowledge** related to environmental sustainability.

1.2.2 Underaged Green Hero (u-GH)

“**Underaged Green Hero**” refers to a Green Hero (GH) who has not reached the minimum legal age required to independently execute binding actions, transactions, or governance participation under applicable laws and regulations.

1.2.3 Visual Representations

a. Mascots

The BE GREEN HEROES (BGH) Program utilizes a defined set of visual mascot representations to symbolically represent a **Green Hero (GH)** within educational, informational, and ecosystem-related materials.

A total of **four mascot images**, comprising **male and female variants**, may be used to represent a Green Hero. These mascots may be displayed independently or in combination with the **Green Hero Coin (GHC)** symbol defined below, solely for purposes of graphic and visual representation of the BGH Program. Such use does not, by itself, imply legal status, eligibility, or participation rights.

Figure 1 depicts Green Hero mascots, **Mia** and **Niko**, intended to represent individuals who are considered **under the legal age** as defined by the applicable laws and regulations of each jurisdiction in which the BGH Program is accessed or referenced. Use of these underage Green Hero (“u-GH”) mascots may be subject to jurisdiction-specific legal requirements, including consent obligations, and shall not be interpreted as conferring any financial, governance, or execution rights.

Fig. 1: Underage Green Hero (u-GH) Mascots – Mia and Niko



Figure 2 depicts Green Hero mascots, **Elio** and **Nina**, intended to represent individuals who have reached the **legal age of majority** under the applicable laws of each relevant jurisdiction. These mascots represent regular Green Heroes (“GH”) and may be used in contexts consistent with lawful participation in the BGH Program, subject to the terms, conditions, and limitations set forth in this White Paper and related governance documents.

Fig. 2: Green Hero (GH) Mascots – Elio and Nina



All mascot representations are symbolic in nature and are used exclusively for branding, educational, and illustrative purposes within the BGH ecosystem.

b. **BGH Symbol (Non-Token)**

Fig.3: BGH Representative Logo



The BGH symbol represents the broader mission and ecosystem identity of the Program. It may be used independently or in supporting visual materials but must not replace the official BGH lettermark in governance, legal, or protocol-level documentation.

c. **BGH Lettermarks**

Fig.4: Two-color Version



The BGH lettermark represents the core identity of the BE GREEN HEROES program. The use of green and blue reflects the dual foundation of the initiative: environmental responsibility grounded in scientific rigor, governance integrity, and long-term systemic thinking.

Green symbolizes learning, growth, and sustainability engagement, while blue represents trust, knowledge, technology, and governance stability. The combined color scheme visually reinforces the Program's commitment to science-based education and credible, decentralized decision-making.

Fig.5: Single-color Version



A single-color version of the BGH lettermark is used in contexts where simplicity, technical clarity, or medium constraints require a monochrome representation.

The single-color version preserves the same geometry and proportions as the primary mark and is intended for use in token icons, governance interfaces, print-limited environments, and other functional applications, while maintaining full brand consistency.

Two mascot images (male and female versions of Green Hero) may be used to represent a Green Hero (GH). Such mascots may also be used together with the **Green Hero Coin (GHC)** symbol defined below for graphic and visual representation of the BGH Program.

Fig.6: Letter-only Version



The letter-only two-color version of the BGH lettermark combines green and blue within a single, simplified typographic mark, without additional symbols or graphic elements.

This version preserves the symbolic meaning of the primary color system—green representing sustainability, learning, and engagement, and blue representing science, governance, and long-term stability—while prioritizing visual clarity and adaptability across digital, technical, and compact use cases. It is intended for applications where a balanced expression of mission and governance is desired within a minimal, letter-focused format, while maintaining full brand consistency.

- d. A separate document, titled as **Brand Assets and Usage Guidelines** (Document code: BGH-EX-RF-0005-2026-EN) is available for more detailed information.

1.3 Green Hero Coin (GHC)

1.3.1 GHC

“**Green Hero Coin**” or “**GHC**” refers to the tokenized cryptocurrency unit within the BGH ecosystem and is denominated in nominal units of **1 GHC**, with an increment value of **1**.

1.3.2 Token-Optimized Identity (Market Context)

The BGH Token Symbol (**Fig. 7**) represents the Token-Optimized Identity of Green Hero Coin (GHC).

This identity is used exclusively in blockchain-facing, exchange-facing, wallet, and token-identification environments where standardized circular token representation is required.

The Token-Optimized Identity is designed for technical clarity, market recognition, and infrastructure compatibility.

Fig. 7: BGH Token Symbol (Token-Optimized Circular Logo)



The minimum divisible unit of GHC is **0.00000000001 GHC (10^{-10} GHC)** and is referred to as **1 Sae**.

1.3.3 Ecosystem Coin Identity (Learning and Governance Representation)

In addition to its Token-Optimized Identity used for blockchain and market infrastructure environments, Green Hero Coin (GHC) also operates under an Ecosystem Coin Identity layer.

The Ecosystem Coin Identity is a circular coin representation designed to reflect the learning-centered and governance-driven nature of the BGH Program. While the Token-Optimized Identity emphasizes market compatibility and technical recognition, the Ecosystem Coin Identity emphasizes:

- knowledge-based participation

- governance eligibility
- lifecycle classification
- structured ecosystem roles

The Ecosystem Coin Identity may incorporate visual elements symbolizing education, learning progression, and governance architecture. It is used in:

- governance documentation
- protocol diagrams
- dashboard interfaces
- educational materials
- ecosystem architecture overviews

The Ecosystem Coin Identity does not replace the Token-Optimized Identity in blockchain explorers, exchanges, wallets, or token contract environments. Instead, it provides internal structural clarity within the BGH ecosystem.

Fig.8: BGH Ecosystem Coin Symbol (Learning and Governance Representation)



1.4 Sub-categorization of Green Hero Coin (GHC)

Under the Ecosystem Coin Identity layer described in Section 1.3.3, Green Hero Coin (GHC) is categorized into functional sub-types to define issuance methods, governance eligibility, lifecycle state, and transferability.

These classifications operate within the internal governance and ecosystem architecture of BGH and are visually represented using the approved Ecosystem Coin Identity circular representations.

Fig. 9: Ecosystem Coin Identity – UC & MC Classifications



1.4.1 Utility Coin (UC)

‘Utility Coin’ or ‘UC’ consists of the following sub-categories of GHC:

- **Mined Utility Coin (m-UC)**
- **Virtual Utility Coin (v-UC)**
- **Provisional Utility Coin (p-UC)**

1.4.1.1 Mined Utility Coin (m-UC)

“**Mined Utility Coin**” or “**m-UC**” refers to GHC designated to be earned by a Green Hero (GH).

- a. The activity of gaining GHC through Learning and Engagement within the BGH Platform is referred to as “**Mining.**”
- b. Immediately following Mining, m-UC becomes **Locked** as part of UC.
- c. m-UC remains **Locked** until the holder performs a specific action described in **Section 1.4.3 (Unlocking)**.

1.4.1.2 Virtual Utility Coin (v-UC)

“**Virtual Utility Coin**” or “**v-UC**” refers to an abstract coin entity and is used for **Governance** of the BGH Program.

1.4.1.3 Provisional Utility Coin (p-UC)

“Provisional Utility Coin” or **“p-UC”** refers to a provisional classification of Utility Coin (UC) earned by an **Underaged Green Hero (u-GH)** through Learning and Engagement activities within the BGH Platform.

p-UC represents a provisional form of Mined Utility Coin (m-UC) and is recorded in recognition of participation, learning progress, and achievement within the Platform.

The functional attributes, rights, restrictions, and protocol-level treatment of p-UC are

1.4.1.3.1 Non-Execution Period (NEP)

“Non-Execution Period” refers to the period during which Provisional Utility Coin (p-UC) remains restricted from execution, unlocking, conversion into economically active tokens, transfer, or any form of commercial or market-related activity.

During the Non-Execution Period:

- p-UC **may be used for governance participation** as defined in **Section 1.4.1.3(c)**;
- p-UC **shall not be used for commercial purposes**, including trading, exchange, monetization, or financial settlement.

The **Non-Execution Period (NEP)** applies until the Underaged Green Hero reaches the minimum legal age required for independent commercial participation, or until such other conditions as defined by protocol rules or applicable laws are satisfied.

1.4.2 Market Coin (MC)

“Market Coin” or **“MC”** refers to GHC designated for trading, transferring, or exchanging in markets (where applicable), including GHC that is issued by the BGH Organization and/or unlocked from m-UC.

1.4.2.1 MC Issued by the BGH Organization (MC-original / MC-o)

MC refers to GHC issued by the BGH Organization for trading, transferring, or exchanging, and which remains permanently tradable in the market.

- a. The activity to issue GHC for trading, transferring, or exchanging is referred to as **“MC Release.”**
- b. This category may also be referred to as **MC-original (MC-o)**.

1.4.2.2 Unlocked m-UC Becomes MC-converted (MC-c)

GHC initially mined by a Green Hero (GH) (i.e., **m-UC**) and subsequently **Unlocked** is referred to as **MC-converted (MC-c)**, regardless of whether it remains in the holder’s wallet, is traded, exchanged, or transferred.

1.4.3 Unlocking

“Unlocking” refers to the specific action taken by a GHC holder to unlock m-UC from its locked Utility Coin (UC) status with or without intention to trade, transfer or exchange. Further details on **“Unlocking”** are provided in **Section 9. Tokenomics**.

Ecosystem Coin Identity (Non-Token-Optimized Perspective)

The circular coin representations shown above constitute the Ecosystem Coin Identity layer of GHC.

These visual identities are used to represent functional, governance, and lifecycle distinctions within the BGH ecosystem, including Utility Coin (UC) and Market Coin (MC) classifications.

The Ecosystem Coin Identity does not replace the Token-Optimized Identity used in blockchain and exchange environments. Instead, it provides structural clarity within governance documentation, dashboards, educational materials, and protocol-level visualizations.

The following coin types may be represented using the approved circular logos shown above:

- **UC:** Utility Coin
- **m-UC:** Mined Utility Coin
- **p-UC:** Provisional Utility Coin
- **v-UC:** Virtual Utility Coin
- **MC:** Market Coin
- **MC-c:** Market Coin-converted
- **MC-o:** Market Coin-original

1.5 Market Coin Release (MCR)

1.5.1 MC Release

“MC Release” refers to an action taken by the **BGH Organization** to release **Market Coin (MC)** into the market for trading, transferring, or exchanging.

1.5.2 Treasury Protocols

MC Release shall follow treasury protocols and procedures established under the **Tokenomics** framework of this White Paper

1.6 Treasury

“Treasury” refers to the primary financial governance tool and mechanism used to securely hold and manage digital assets of the BGH ecosystem, particularly Market Coin (MC). Further details on Treasury and Treasury Protocol(s) are provided in **Section 10**.

1.7 Governance, Voting, and Voting Power (VPo)

1.7.1 Governance

“Governance” refers to any and all activities and decisions required to manage the operations of the BGH Program and the BGH Organization.

1.7.2 Voting

“Voting” refers to the act of casting a vote on a Proposal posted on the BGH Platform.

1.7.3 Voting Power (VPo)

“Voting Power” or **“VPo”** refers to the eligibility and unit value for voting within the BGH Platform, and is granted to Utility Coin (UC) holders.

1.8 Learning and Engagement Content

1.8.1 Content Definition

“Learning and Engagement Content” refers to all content and information, whether educational or promotional, to be published on the BGH Platform.

1.8.2 Author

“Author” refers to the person who owns the copyright(s) of content that may be proposed and voted for inclusion in the BGH Platform.

1.8.3 Proposer

“Proposer” refers to a person who may or may not be the Author, and who proposes content for voting and potential inclusion in the BGH Platform. This term applies to all other clauses in this document, where **Proposal** and **Voting** are relevant

1.9 Legal and General White Paper Definitions (Standard)

1.9.1 White Paper

“White Paper” refers to this document and any revisions, appendices, or referenced materials released by the **BGH Organization** in connection with the **BGH** ecosystem and **GHC**.

1.9.2 Not an Offer / No Solicitation

“Not an Offer” means that this White Paper is provided for informational purposes only and does not constitute an offer to sell, or solicitation of an offer to buy, any securities, financial instruments, or investments in any jurisdiction.

1.9.3 No Investment Advice

“No Investment Advice” means that nothing in this White Paper shall be interpreted as financial, investment, legal, tax, or trading advice.

1.9.4 Restricted Persons

“Restricted Persons” refers to individuals or entities who are prohibited from participating in any token distribution, platform access, or related activity due to applicable laws, sanctions, or regulatory restrictions in their jurisdiction.

1.9.5 Digital Assets

“Digital Assets” refers to cryptographic tokens, cryptocurrencies, stablecoins, NFTs (if applicable), and other blockchain-based or digital representations of value that may be held, transferred, or managed through wallets and/or smart contracts.

1.9.6 Wallet

“Wallet” refers to a software and/or hardware mechanism used to store cryptographic private keys and interact with blockchain addresses used to hold or transfer digital assets, including GHC where applicable.

1.9.7 Smart Contract

“Smart Contract” refers to a blockchain-based program used to automate token-related mechanics and operations, including issuance, transfers, locking/unlocking, governance functions, and/or treasury management where applicable.

1.10 Notes on Definitions and Terminology

Given the scope of content and the introduction of new concepts throughout this White Paper, it may be difficult and inefficient to define every necessary term exclusively within

this Section. Accordingly, additional sections and subsections of this White Paper may introduce further definitions to support comprehensive reader understanding.



Section 2. Purposes and Goals

This section defines the core philosophy and strategic objectives of the BE GREEN HEROES (BGH) Program. These purposes and goals guide the design of the BGH Platform, its learning system, tokenized participation model, and governance framework.

Participation in the BGH Program includes both adult and underaged participants (“Green Heroes”), with learning, engagement, and participation mechanisms structured to support age-appropriate inclusion in accordance with applicable laws and protocol rules.

2.1 Advance Science-Based Education for Environmental Sustainability

2.1.1 The Challenge

Across the world, environmental understanding is frequently shaped by misinformation, oversimplified narratives, and incomplete lifecycle thinking. Many widely repeated sustainability claims lack scientific rigor or fail to account for full environmental trade-offs.

2.1.2 Why It Matters

When weak ideas are repeated often enough, they become accepted as “common sense.” These misconceptions can distort priorities, misdirect funding, and result in sustainability actions that unintentionally harm the environment—even when intentions are positive.

2.1.3 How It Spreads

The spread of misinformation accelerates when influential actors—including governments, international organizations, NGOs, corporations, media, and activists—amplify unverified claims without technical validation. Social media virality, marketing language, and agenda-driven advocacy often outpace careful scientific analysis.

2.1.4 Call to Action

The BGH Program invites the public to participate in a facts-forward culture of environmental stewardship. By elevating science-based education and rigorous lifecycle thinking, the BGH ecosystem aims to redirect global effort toward strategies that generate measurable environmental impact, accelerate credible innovation, and build long-term public trust in sustainability decision-making.

2.2 Launch BGH with an Initial Focus on Textiles, Then Expand

The BGH Program is designed to launch with an initial thematic focus on textile sustainability, with expansion into other high-impact industries and environmental domains over time.

2.2.1 Why Textiles

Textiles are universally consumed at high volume throughout human life and generate substantial waste. As a result, textiles are relatable to the global population and represent a practical starting point for scalable sustainability education.

2.2.2 The Knowledge Gap

Although textiles are among the major contributors to environmental harm—from raw material cultivation through manufacturing, distribution, consumption, and disposal—most consumers do not understand the full lifecycle impacts of textile products or their own behavioral influence within the system.

2.2.3 Misinformation in Global Discourse

Due to the lack of education and accessible scientific framing, misconceptions commonly shape sustainability discourse and consumer behavior, reinforcing patterns described in Section 2.1.

2.2.4 Structural Education Problem

A major barrier to effective learning is the widespread assumption that people already understand textile sustainability. Familiarity with clothing and daily textile use creates confidence without the scientific foundation required for accurate decision-making.

2.2.5 Core Objective

BGH aims to deliver science-based textile sustainability education globally, enabling individuals to make more favorable decisions in:

- purchasing and consumption behavior
- product care and maintenance habits
- repair, reuse, and disposal practices

2.2.6 Expansion Beyond Textiles

The principles described in this section are intended to apply beyond textiles. After establishing proven educational structures and engagement mechanisms, the BGH Program will expand into additional industries and domains that generate significant environmental impact.

2.3 Incentivize Learning Through Green Hero Coin (GHC)

The BGH Program introduces a token-based participation model designed to incentivize learning and sustained engagement while maintaining alignment with the Program’s educational mission and governance safeguards.

2.3.1 Learning-to-Earn Incentives

Participants (“Green Heroes”) are eligible to earn Green Hero Coin (GHC) by completing learning activities and meeting platform-defined requirements.

2.3.2 Potential Economic Benefit

If GHC gains market value over time, eligible participants may benefit financially, subject to applicable protocol rules, participation status, and legal eligibility. This incentive mechanism is intended to support sustained learning engagement and long-term program growth while preserving governance integrity and mission alignment.

2.4 Operate Under a Non-Profit Business Model

BGH is structured as a non-profit initiative designed to maintain independence, transparency, and scientific integrity.

2.4.1 Mission-Centered Funding

The BGH Organization raises funds and reinvests them into science-based environmental education, platform development, and global learning initiatives.

2.4.2 Independence and Integrity

To remain free from external influence and agenda-driven pressure, the BGH Organization is designed to operate under a non-profit business model.

2.4.3 Self-Sustaining Revenue Strategy

Unlike conventional non-profits that rely heavily on donations or government and corporate funding, BGH is designed to rely primarily on internal revenue streams during the The Execution Period (TEP), including:

- Market Coin (MC) release mechanisms
- subscription-based participation models

2.4.4 Long-Term Funding Flexibility

After the TEP, the Operating Committee may evaluate whether external funding sources (e.g., partnerships or sponsorships) should be pursued to ensure continued long-term sustainability.

2.5 Milestones During the The Execution Period (TEP)

2.5.1 Milestone 1: Global Learning Participation

The first milestone is to achieve broad public participation such that a significant portion of the global population engages in the BGH learning platform and earns Utility Coin through participation.

Mining Principle 1: Participation Concentration

The platform may allow individual users to earn multiple mined tokens within defined limits. While this may result in uneven distribution across the population, it is intentionally designed to support network effects, whereby highly engaged participants amplify learning within their communities and accelerate broader reach.

Mining Principle 2: Rate Limits

Mining follows time-based limits (daily, weekly, monthly, and annual), consistent with tokenomics design and fairness safeguards.

2.5.2 Milestone 2: Institutional Adoption

The second milestone is to implement BGH textile education content in high school curricula within designated Target Countries (TTC).

Target Countries are selected based on their influence on textile consumption and their potential to amplify educational reach globally.

2.6 Expected Outcomes After Achieving Milestones

If both milestones are achieved during the The Execution Period, the BGH Organization anticipates the following outcomes:

2.6.1 Educational Reach

By achieving large-scale Utility Coin distribution through learning participation, a meaningful global population segment gains validated sustainability knowledge and becomes capable of spreading that knowledge through community influence.

2.6.2 Governance Integrity

As the collective knowledge base strengthens, governance decisions are expected to align more reliably with scientific credibility and the Program's original goals, supporting long-term sustainability beyond the TEP.

2.6.3 Long-Term Operational Sustainability

As participation expands globally, the Program may transition toward additional sustainability strategies after the TEP, including partnerships and other mechanisms supporting long-term execution.

2.6.4 Content Sustainability

While educational content remains subject to governance decisions, widespread science-based learning increases the likelihood that the platform continues to evolve responsibly and retains alignment with its original purpose.

2.7 Utilize an Energy-Efficient Blockchain Infrastructure

BGH incorporates blockchain technology to enable transparent rule enforcement, programmable token distribution, and decentralized governance mechanisms.

2.7.1 Sustainability Consideration

Although blockchain technology is often criticized for energy use, BGH prioritizes platforms where environmental operating costs are reduced and where the educational and governance benefits justify the infrastructure choice.

2.7.2 Internet Computer Protocol (ICP)

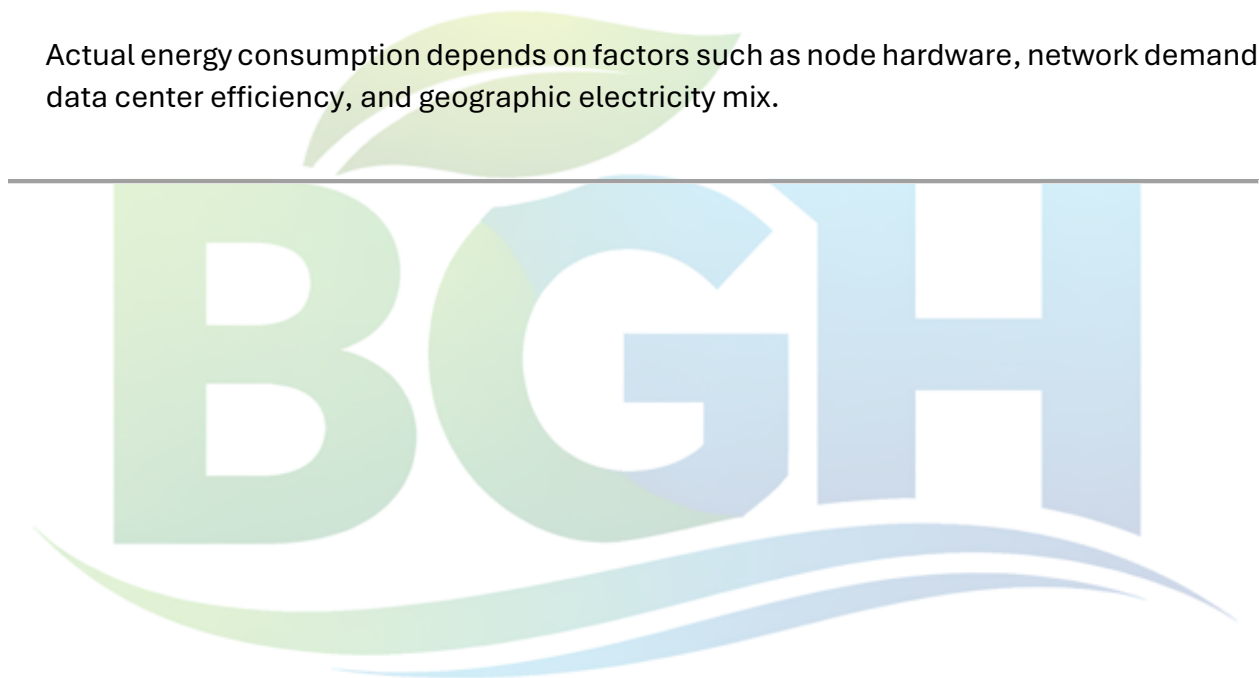
At launch, BGH deploys its on-chain infrastructure on the Internet Computer Protocol (ICP), selected for its energy-efficient architecture relative to proof-of-work (PoW) networks.

Unlike PoW blockchains that rely on energy-intensive mining competition, ICP achieves security without PoW mining and avoids the primary source of electricity consumption associated with PoW systems.

Further details are provided in a separate document, *Environmental Impact & Energy Efficiency* (Document code: BGH-RF-0006-2026-EN).

2.7.3 Real-World Variables

Actual energy consumption depends on factors such as node hardware, network demand, data center efficiency, and geographic electricity mix.



Section 3. Learning and Engagement

This section outlines the Learning and Engagement pillar of the BGH Platform, designed to educate participants, strengthen public involvement, and enable users to earn Green Hero Coin (GHC), or its provisional equivalent where applicable, through meaningful sustainability-focused activities aligned with the goals described in Section 2.

3.1 Definition

Learning and Engagement refers to the full range of educational content, public-facing learning tools, and participation-driven activities provided or endorsed by the BGH Organization to support sustainability awareness, scientific literacy, and informed environmental action.

3.2 Content Types

Content is considered eligible for inclusion on the BGH Platform where it is deemed beneficial for Green Heroes (GHs), including both adult and underaged participants, and is proposed and approved through the applicable governance processes.

Eligible content may include, but is not limited to:

- books, articles, and research papers
- blogs, websites, and digital learning resources
- events and other educational or promotional materials

3.3 Operational Framework

3.3.1 Initial Learning Content at Official Public Launch (OPL)

At the time of the Official Public Launch (OPL), the following works are incorporated into the Learning and Engagement components of the BGH Platform:

- *BE GREEN HEROES*, written by Sae Chang
- *COLD WINTER WARM WINTER*, written by Sae Chang
- *Environmental Negligence and Procrastination*, co-written by Ingrid Mellado Hoyos and Sae Chang

3.3.2 International Sustainability Marketing Competition (ISMC)

The BGH Organization assumes responsibility for organizing the International Sustainability Marketing Competition (ISMC), previously managed by Clean Recycling Initiative™ and HEAT-MX™.

ISMC serves as a primary engagement mechanism at and following OPL, enabling participants to engage in sustainability-focused learning activities and to earn GHC, or its provisional equivalent where applicable, through structured, mission-aligned participation.

3.3.3 Expansion of Learning and Engagement Resources

The BGH Organization will continuously expand Learning and Engagement opportunities on the Platform, which may include additional books, research publications, digital resources, event programming, announcements, and other mission-relevant materials.

All new additions follow the governance mechanisms described in Section 7 (Proposal) and Section 8 (Voting).

3.3.4 Updates and Modifications

Any updates to existing Learning and Engagement content, including additions, removals, or revisions, shall be governed by the same governance mechanisms described in Section 7 (Proposal) and Section 8 (Voting).

Section 4. Stakeholders

This section defines the stakeholder structure of the **BGH Organization**, including founding member and governance entities that support execution and oversight of the **BGH Program** at and after the **Official Public Launch (OPL)**.

4.1 Founding Member

The **Founding Member** is the individual who established and created the BGH Program:

- **Sae Chang**

4.2 Operational Committee(OC)

4.2.1 Definition

The **Operating Committee (OC)** serves as the primary execution and management body of the BGH Organization and is responsible for operational delivery of the BGH Program.

4.2.2 Composition

The OC consists of individuals who are hired or elected to manage day-to-day operations. Members may include, where appropriate, individuals who also serve on the Board.

4.2.3 Compensation

Compensation for OC members will be funded through proceeds generated via the **ICO** and **MC Release**, in accordance with the **Treasury Protocol (TTP)**.

4.3 Board

4.3.1 Formation and Compliance

The BGH Organization will form a Board of Directors in compliance with applicable Canadian non-profit incorporation laws and other relevant regulatory requirements aligned with its operational model.

4.3.2 Role and Responsibilities

The Board is responsible for making key organizational decisions, including matters that directly or indirectly affect the goals and purposes defined in **Section 2**, and those customary to the oversight function of a nonprofit organization.

4.3.3 Compensation

Board compensation follows the same principle defined for the Operating Committee under **Section 4.2.3**, and is subject to the Treasury Protocol (TTP).



Section 5. Official Public Launch (OPL) and Immutability

This section defines the **Official Public Launch (OPL)** of the BGH Program and the framework governing the transition to **smart contract immutability** for the BGH Platform's on-chain components.

5.1 Official Public Launch (OPL)

5.1.1 Definition

The **Official Public Launch (OPL)** is the public commencement of the BGH Program, marked by an official announcement enabling users to begin utilizing the BGH Platform—particularly the Learning and Engagement system and the earning and mining of **Green Hero Coin (GHC)**.

5.1.2 Launch Communications

The launch is communicated through formal and public-facing channels, including:

- Press releases
- Official social media publications

Launch communications will be distributed across all jurisdictions where participation and operations are legally permitted and consistent with the ethical standards of the BGH Organization.

5.1.3 Operational Readiness at Launch

At **OPL**, the **BGH Organization** confirms that:

- The core on-chain and platform infrastructure is operational and accessible
- The user experience supports intuitive participation for Green Heroes (GHs)
- The foundational program structure is deployed, with only limited refinements remaining for post-launch stabilization

5.2 The Grace Period (TGP)

5.2.1 Definition

The **Grace Period (TGP)** is a defined stabilization window following OPL and prior to full smart contract immutability. Its purpose is to allow the BGH Organization to evaluate early platform operations and ensure readiness for permanent on-chain governance and execution.

5.2.2 Duration

The TGP is a maximum period of **18 months** from OPL, after which smart contract immutability must be implemented.

5.2.3 Scope of Adjustments During TGP

During TGP, the BGH Organization may implement improvements to the BGH Platform's on-chain execution, provided these adjustments do not alter the fundamental scope and principles of the BGH Program.

The following areas remain protected from material change during TGP:

- The core mission, purposes, and goals described in **Section 2**, including referenced governance objectives
- The governance architecture and governance protocol described in **Section 6**
- Founders' GHC allocations, except where a reduction may be applied (e.g., corrective action or voluntary decrease)

5.3 Immutability

5.3.1 Definition

Immutability refers to the permanent unchangeability of the smart contract(s) governing BGH Platform on-chain behavior, token rules, and tokenomics execution.

5.3.2 Implementation

Following smart contract validation and completion of required stabilization activities during TGP, the BGH Platform's on-chain system will transition to full immutability at the earliest feasible time.

Section 6. Governance and Governance Protocol

This section defines the governance framework of the BGH Program, including its guiding principles, voting eligibility, governance instruments, and decision-making domains that regulate platform evolution and collective action.

6.1 Governance Principles

6.1.1 Decentralized Decision-Making

The BGH Program operates under a decentralization-first model in which platform evolution, policy formation, and strategic direction are governed through structured Proposal and Voting mechanisms executed via the Voting Process (VPr).

6.1.2 Voting Eligibility and Authority

Governance participation within the BGH Program is exercised through the Voting Process (VPr) by eligible participants holding governance-bearing instruments as defined by protocol rules, including:

- Utility Coins (UCs)
- Provisional Utility Coins (p-UC), subject to applicable restrictions
- Voting Power (VPO) assigned in accordance with protocol rules

Governance participation confers authority solely over governance matters and **does not imply or grant economic, commercial, or market-based rights**, unless explicitly stated elsewhere in this White Paper.

6.1.3 Provisional Utility Coin (p-UC) Governance Participation

Notwithstanding its provisional, non-executable, and non-commercial status, **Provisional Utility Coin (p-UC) participates in governance during the applicable Non-Execution Period**, as defined in Section 1.4.1.3.1.

Governance participation by p-UC may include, subject to protocol rules:

- governance signaling and feedback
- review and deliberation of Proposals
- participation in voting processes within defined governance domains

p-UC **does not grant economic rights, commercial privileges, transferability, or market access**. Governance participation by p-UC is strictly limited to educational, civic, and ecosystem-development purposes and is structurally separated from all forms of commercial activity.

6.1.4 Market Coin (MC) Exclusion

Market Coin (MC) is explicitly designated as non-governance-bearing.

Holders of MC are not eligible to submit proposals, participate in governance deliberation, or vote within the BGH Program.

6.1.5 Scientific Integrity and Decision Protection

Governance mechanics, eligibility rules, and voting power distribution are designed to prioritize decisions aligned with credible scientific principles and to mitigate the risk of misinformation, misperception, or economically motivated influence affecting outcomes related to global environmental action.

6.1.6 Virtual Utility Coins (v-UC)

Virtual Utility Coins (v-UC) represent a non-transferable governance allocation exercised by the BGH Organization during early-stage program development.

v-UC participates in Voting Power (VPo) calculations under the Tokenomics model but is structurally designed to regress proportionally as m-UC supply increases.

This mechanism reflects the staged transition from Organization-weighted governance to Community-weighted governance over time.

6.1.7 Gradual Voting Power Redistribution

While the rate at which participants mine Utility Coins is not directly controlled by the BGH Organization, Voting Power distribution is structured to progressively increase governance influence for Green Heroes as their engagement, knowledge, and scientific literacy evolve over time.

A theoretical voting power redistribution model is presented in Section 8.5 (*Theoretical VPo Redistribution Models*).

6.2 Governance Structure

Governance within the BGH Program is organized into the following domains:

6.2.1 Governance – Contents

Governance – Contents applies to decisions involving educational, informational, and promotional materials relevant to the Program’s mission, including content distributed through Learning and Engagement activities.

This domain governs decisions including, but not limited to:

- adoption of new content into the Platform
- removal of outdated, inaccurate, or unsuitable content
- modification, revision, or enhancement of Learning and Engagement materials

All key actions within this domain are executed through the Voting Process (VPr).

6.2.2 Governance – Operations

Governance – Operations applies to all operational, procedural, and organizational decisions not included within Governance – Contents, including actions required to maintain, adapt, and evolve the BGH Platform.

All key actions within this domain are executed through the Voting Process (VPr).

6.3 Governance Participation and Commercial Separation

Participation in governance within the BGH Program, including participation by holders of Provisional Utility Coin (p-UC), **does not constitute commercial activity** and shall not be construed as granting rights to trade, monetize, exchange, or otherwise engage in market-based actions.

Governance participation is structurally, technically, and legally separated from:

- token execution and transferability
- market access or exchange activity
- financial settlement or monetization

Commercial participation, where permitted, is governed exclusively by separate protocol rules, eligibility conditions, and applicable laws.

Section 7. Proposal and Proposal Protocol

This section defines the Proposal mechanism and its governing protocol, which serve as the primary entry point for governance-driven changes to the BGH Program, the BGH Platform, and the BGH Organization.

7.1 Definitions

7.1.1 Proposal

A *Proposal* is a formal submission requesting a change, addition, or adjustment to eligible governance structures, protocols, content, or operational components of the BGH ecosystem.

7.1.2 Scope

Proposals may address any eligible aspect of BGH governance or operations, including but not limited to:

- **Governance – Contents**, such as Learning and Engagement content additions, removals, or revisions
- **Governance – Operations**, such as operational procedures, platform workflows, and administrative execution

7.1.3 Proposer

A *Proposer* is the individual or entity submitting a Proposal in accordance with the eligibility requirements defined in this Section.

7.1.4 Support

Support refers to the act of endorsing a Proposal to qualify it for progression into the Voting Process (VPr), in accordance with the thresholds defined herein.

7.1.5 Proposal Status

Proposals progress through the following status stages:

- **Proposal–Initial:** Submitted and collecting required Support
- **Proposal–Supported:** Meets Support requirements and advances to the Voting Process

7.2 Governance Principles for Proposals

7.2.1 Purpose

The Proposal Protocol is designed to enable efficient, community-led governance while maintaining the stability, scientific integrity, and mission alignment of the BGH Program.

7.2.2 Eligibility to Submit Proposals

Any participant holding more than **150 Utility Coins (UCs)** is eligible to submit a Proposal, subject to protocol rules.

For the purposes of Proposal eligibility, Utility Coins may include:

- Mined Utility Coins (m-UCs)
- Provisional Utility Coins (p-UCs), subject to applicable restrictions
- Virtual Utility Coins (v-UCs), where assigned

Proposal eligibility confers **governance participation only** and does not grant economic, commercial, or market-based rights.

7.2.3 No Automatic Adoption

Submission of a Proposal does not guarantee implementation. All Proposals must satisfy eligibility, Support, and Voting requirements before adoption.

7.2.4 Immutable vs. Adjustable Governance Areas

To protect the foundational mission, structure, and integrity of the BGH Program, certain areas are designated as immutable and are not eligible for modification through Proposals.

(a) Immutable Areas (Not Eligible for Proposals)

Immutable areas include, but are not limited to:

- Core principles, mission, and foundational values of the BGH Program
- Total token issuance quantity
- Allocation ratios between Market Coin (MC) and Utility Coin (UC)
- Fundamental mechanisms governing learning-based earning of Utility Coins
- Decentralization architecture and Voting Power (VPo) transfer logic

- Organizational names and identity, including *BE GREEN HEROES (BGH)*
- Non-profit structural framework and foundational operating model
- Core income structures, including subscription model design, fee logic, and charging structure
- Foundational operational protocols essential to system integrity
- Operations of affiliated organizations, including Clean Recycling Initiative™ and HEAT-MX™

(b) Adjustable Areas (Eligible for Proposals)

All other operational metrics explicitly designated as adjustable in this White Paper (marked with the superscript “o”) may be proposed for modification.

7.3 Proposal Protocol

7.3.1 Submission Tools

All Proposals must be submitted using the official Proposal tools provided within the BGH Platform.

7.3.2 Content Ownership and Copyright

If the Proposer is not the author or rights holder of the content being introduced or modified, the Proposer must confirm that appropriate permissions have been obtained from the copyright owner.

By submitting a Proposal, the Proposer confirms the legitimacy of the submission and agrees to the Platform’s applicable terms.

Any disputes, claims, or legal actions related to copyright infringement remain the sole responsibility of the Proposer.

7.3.3 Minimum Support Threshold

To advance to the Voting stage, a Proposal must:

- Secure Support from at least **two (2)** other Utility Coin holders; and
- Reach a minimum combined Support threshold equivalent to:

$$150 \times 100 = 15,000 \text{ UCs}$$

Support may be provided by any combination of eligible UC holders, subject to protocol rules.

7.3.4 Support Time Limit

A Proposal must meet the minimum Support threshold within **7 days** of submission. If the threshold is not met within this timeframe, the Proposal is closed as not qualified for Voting.

7.3.5 Re-submission Window

A Proposal that fails to meet the minimum Support threshold may only be re-submitted after a minimum waiting period of **six (6) months**.

7.3.6 Voting Eligibility

Only a **Proposal-Supported** may proceed to the Voting Process (VPr).

7.3.7 Final Approval Requirement

Progression to the Voting Process does not constitute final approval. A Proposal is adopted only if it successfully completes the Voting Process and meets all approval requirements defined in the Voting Protocol.

Section 8. Voting and Voting Protocol (VPt)

This section defines the Voting Process (VPr) and the Voting Protocol (VPt), through which the BGH community approves or rejects eligible Proposals within a transparent, decentralized governance framework.

8.1 Definitions and Overview

8.1.1 Voting as the Primary Governance Mechanism

As a decentralized program, the BGH Program executes key governance decisions through a standardized Voting Process (VPr), governed by the Voting Protocol (VPt) described in this section.

8.1.2 Voting Eligibility

Any holder of governance-bearing Utility Coins (UCs) is eligible to participate in voting on Proposals posted for voting on the BGH Platform, subject to protocol rules.

Eligible Utility Coin categories may include:

- Mined Utility Coins (m-UCs)
- Provisional Utility Coins (p-UCs), subject to applicable restrictions
- Virtual Utility Coins (v-UCs), where assigned

Voting eligibility confers **governance participation only** and does not imply economic, commercial, or market-based rights.

8.1.3 Non-Eligibility of Market Coin (MC)

Market Coin (MC) does not carry Voting Power (VPo). Holders of MC are not eligible to participate in governance voting.

8.1.4 Voting Power (VPo)

Voting within the BGH Program is weighted using Voting Power (VPo), calculated in nominal units and applied consistently across all governance decisions in accordance with the Voting Protocol.

8.1.5 Proposal Status Outcomes

Following completion of the Voting Process:

- A Proposal that passes voting is designated **Proposal–Approved (Pro-A)**
- A Proposal that fails voting is designated **Proposal–Rejected (Pro-R)**

8.2 Governance Principle 1: “Not-To-Do” Safeguards

The voting and governance design of the BGH Program is explicitly structured to prevent adoption of approaches that reflect common failure patterns observed in global sustainability initiatives.

These include, but are not limited to:

- misinformation, distortion, and public misperception
- claims based on weak or non-rigorous science
- politically or commercially driven environmental agendas
- oversimplified messaging lacking context, trade-off analysis, or evidence
- narrowly focused solutions that ignore broader systemic drivers
- greenwashing and symbolic actions without measurable outcomes
- end-of-pipe solutions that ignore upstream causes
- technology-only narratives that exclude behavioral and institutional change
- one-size-fits-all policies that ignore local realities and justice considerations
- weak monitoring, lack of metrics, and poor transparency
- short-term decision-making and unstable policy cycles
- fragmented governance and cross-sector misalignment
- rebound effects and unintended impacts left unaccounted
- exclusion of affected communities and stakeholders
- overreliance on voluntary commitments without accountability

8.3 Governance Principle 2: Gradual Transfer of Voting Power (VPo)

8.3.1 Early-Stage Voting Power Stability

During the initial phase of the Program, the BGH Organization retains sufficient Voting Power (VPo) to preserve scientific integrity, governance stability, and mission alignment while collective platform knowledge is still developing.

8.3.2 Knowledge-Based Decentralization Over Time

As Green Heroes strengthen their knowledge base through Learning and Engagement and earn m-UCs, the system is designed to gradually transfer Voting Power (VPO) toward the broader community in a controlled and knowledge-weighted manner.

8.4 Voting Process (VPr)

8.4.1 Voting Interface

Once a Proposal reaches the Voting stage, it is published on the BGH Platform dashboard and made available for community voting.

8.4.2 Voting Period

Each voting window remains open for **two (2) weeks** from the published voting start date.

8.4.3 Operating Committee (OC) Participation Requirement

For a voting outcome to be valid, at least **one (1) Operating Committee (OC) member** must cast a vote.

This requirement serves as an early-phase safeguard to reduce the risk of harmful or scientifically unsound proposals passing while platform-wide expertise is still maturing.

8.4.4 Approval Threshold at Launch

At the Official Public Launch (OPL), the minimum approval threshold is set at **30% of the total existing m-UC Voting Power**, to encourage early participation and governance activation.

This threshold may be increased to **50% + 1** when the Program determines that stronger governance rigor is required.

8.4.5 Threshold Calculation

The required Voting Power threshold for each Proposal is calculated and displayed when the Proposal enters the Voting stage, using a standardized time-frame methodology defined by the Voting Protocol.

8.4.6 Post-Vote Notification

Within **seven (7) days** after the voting window closes, the Proposer is notified of the final outcome:

- Approved → **Proposal–Approved (Pro-A)**
- Rejected → **Proposal–Rejected (Pro-R)**

8.4.7 Implementation and Integration

Approved Proposals may require varying integration timelines depending on complexity and urgency.

Some actions may be executed automatically via smart contracts, while others may require structured implementation planning by the BGH Organization. Event-based Proposals may require immediate execution, whereas system-level changes may require staged deployment.

8.5 Theoretical Voting Power (VPo) Redistribution Models

8.5.1 Eligible Voting Power Sources

Voting Power (VPo) is associated with eligible Utility Coin categories, including v-UCs and eligible locked Utility Coin positions, where applicable.

8.5.2 Virtual Utility Coin (v-UC) Supply Logic

The quantity of Virtual Utility Coins (v-UCs) is defined as:

$$\mathbf{v\text{-}UC = 4.75B - \text{Mined Utility Coin (m\text{-}UC)}}$$

This formula operationalizes the Governance Transition Principle, whereby ecosystem-derived Utility Coins progressively replace Organization-reserved governance weight.

As m-UCs are mined over time, the v-UC quantity decreases accordingly.

8.5.3 Voting Power Loss Through Market Conversion

When m-UCs are converted into Market Coin (MC), those units no longer carry Voting Power and are excluded from governance participation.

8.5.4 Voting Power (VPo) Formula

Based on the above relationships:

$$\mathbf{\text{Total VPo} = \text{v\text{-}UC} (\times 1) + \text{m\text{-}UC} (\times 1)}$$

8.5.5 Model Illustrations

Theoretical redistribution models are illustrated in:

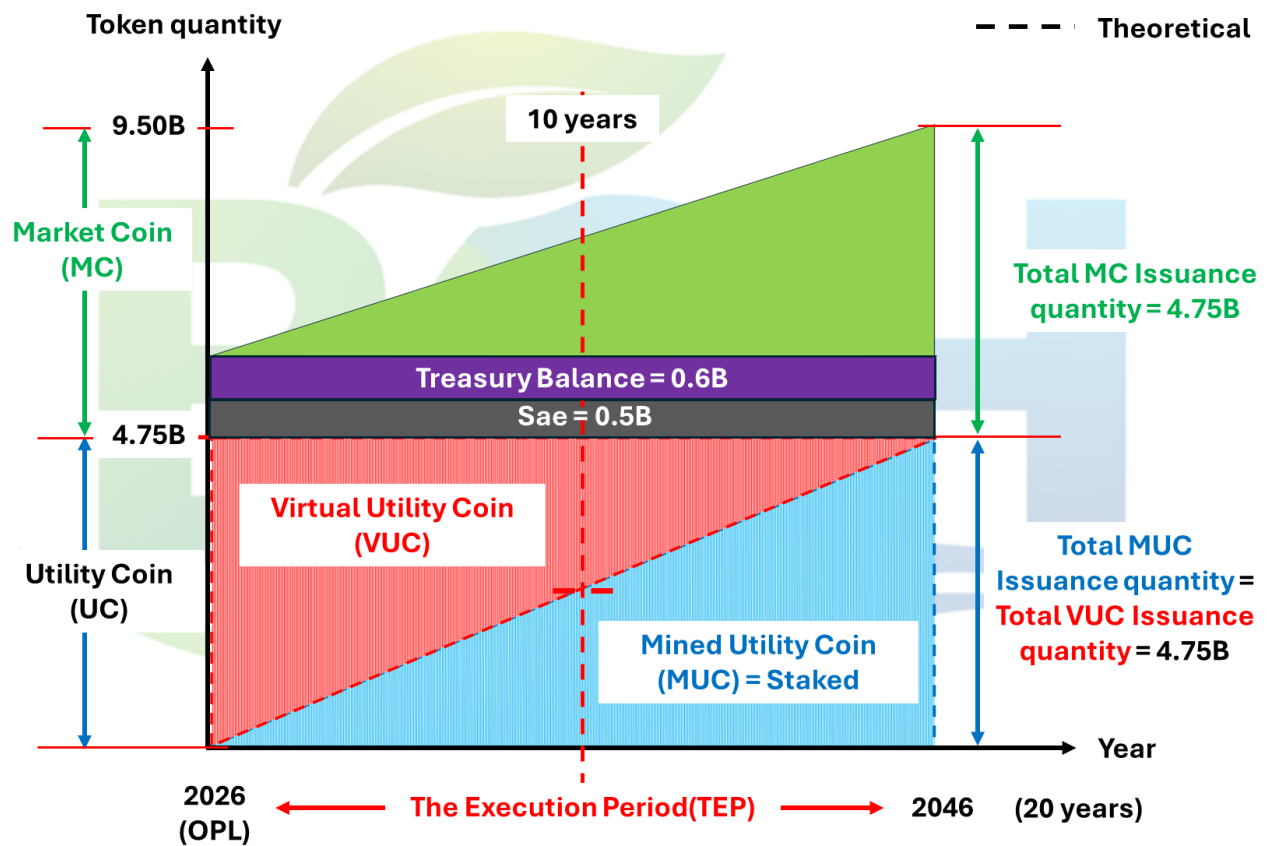
- **Figure 3:** Voting Power redistribution over time
- **Figure 4:** Deflationary Voting Power redistribution over time

8.5.6 Market-Side Effects

m-UCs sold into the market and converted into MC reduce community governance weight and may introduce inflationary pressure on market behavior.

The balance of Voting Power between the BGH Organization and the public is evaluated across multiple modeled scenarios to ensure long-term governance stability.

Fig.10: Tokenomics Overview with Voting Power Distribution Principle



8.6 v-UC VPo Regression Models in 6 Scenarios

8.6.1 Scenario 1 ~ 3

8.6.1.1 1million (1M) m-UCs mined in Year 1 with the following variances

- Annual increase rate (**Scenario 1**) = 10%
- Annual increase rate (**Scenario 2**) = 25%
- Annual increase rate (**Scenario 3**) = 50%

8.6.1.2 **m-UCs converted to MCs at 50% rate**

8.6.1.3 **Total v-UC and Remaining m-UC quantity breakeven point**

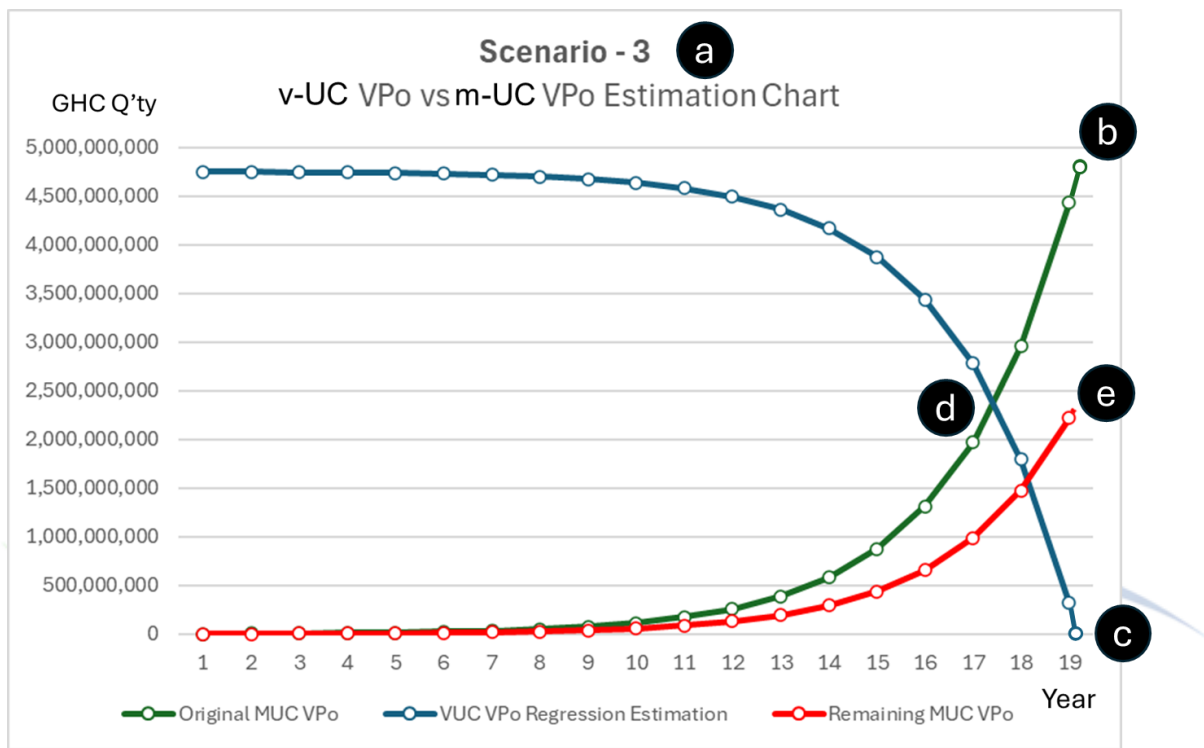
8.6.1.3.1 **Scenario 1 = Total m-UC never reaches v-UC's VPo during The Execution Period (TEP)**

8.6.1.3.2 **Scenario 2 = Same as 8.6.1.3.1**

8.6.1.3.3 **Scenario 3 =**

- a. The **Original m-UC VPo** reaches the **v-UC VPo** around **Year 18.5**
- b. The **Remaining m-UC VPo** reaches the **v-UC VPo** shortly after **Year 18**

Fig.11: v-UC and m-UC Voting Power (VPo) Voting Power Estimation – Scenario 3



- a** Voting Power (VPo) contributions from m-UCs in Scenario 1 & 2 negligible, thus v-UC's VPo remains the dominant force – Graphs for Scenario 1 & 2 not plotted due to the negligence
- b** Maximum v-UC VPo reached shortly after Year 19
- c** V-UC VPo reached to 0 shortly after Year 19
- d** v-UC vs. Original m-UC VPo equal balance (cross) point reached around Year 17.5
- e** v-UC vs. Remaining m-UC equal balance (cross) point reached shortly Year 18

8.6.2 Scenario 4 ~ 6

8.6.2.1 **5million (5M) m-UCs** mined in **Year 1** with the annual increase rate in the same variances as 8.6.1.1.1 for **Scenario 4, 5 and 6** respectively

8.6.2.2 **m-UCs** converted to **MCs** at **50% rate**

8.6.2.3 **Total v-UC and Remaining m-UC quantity breakeven point**

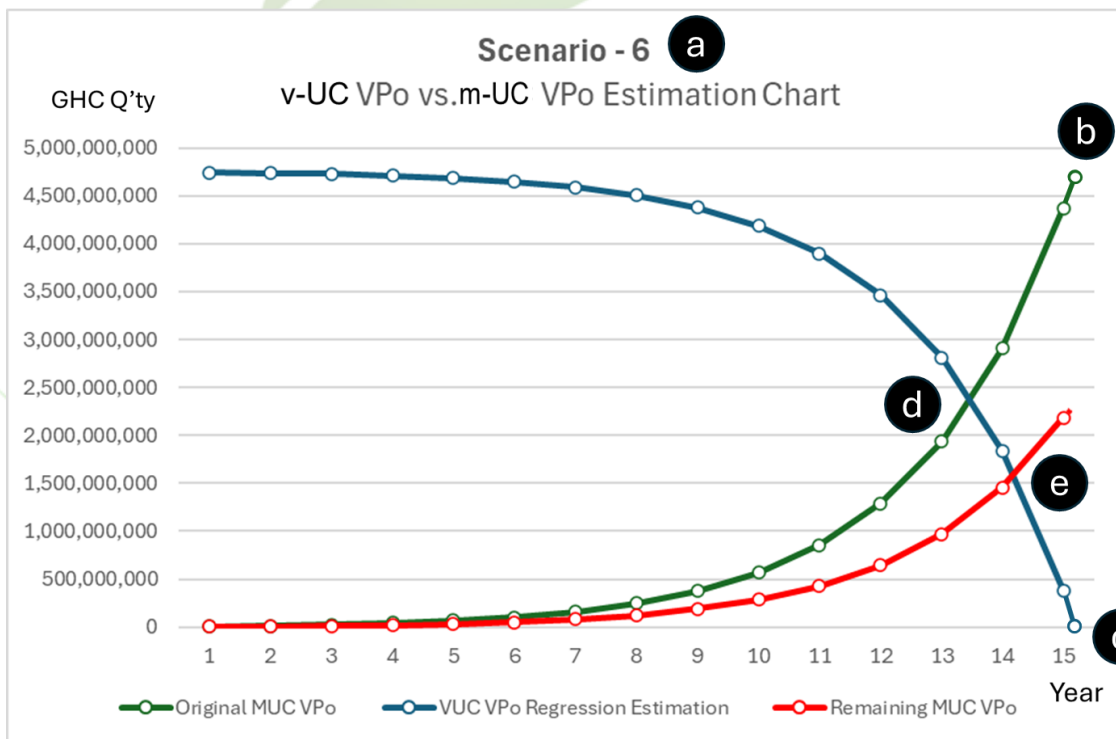
8.6.2.3.1 **Scenario 4 = Total m-UC never reaches v-UC's VPo during The Execution Period (TEP)**

8.6.2.3.2 **Scenario 5 = Same as 8.6.2.3.1**

8.6.2.3.3 **Scenario 6 =**

- The **Original m-UC VPo** reaches the **v-UC VPo** around **Year 13.5**
- The **Remaining m-UC VPo** reaches the **v-UC VPo** shortly after **Year 14**

Fig.12: v-UC and m-UC Voting Power (VPo) Voting Power Estimation – Scenario 6



- Voting Power (VPo) contributions from MUCs in Scenario 1 & 2 negligible, thus VUC's VPo remains the dominant force – Graphs for Scenario 1 & 2 not plotted due to the negligence
- Maximum MUC VPo reached shortly after Year 15
- VUC VPo reached to 0 shortly after Year 15
- VUC vs. Original MUC VPo equal balance (cross) point reached around Year 13.5
- VUC vs. Remaining MUC equal balance (cross) point reached shortly Year 14

Section 9. Tokenomics

This section defines the tokenomics of Green Hero Coin (GHC), including issuance, classification, earning mechanics, governance alignment, release controls, and redistribution dynamics across the BGH ecosystem.

9.1 Token Model Overview

Green Hero Coin (GHC) operates under a dual-layer identity architecture:

1. **Token-Optimized Identity** – used in blockchain, exchange, wallet, and market infrastructure environments.
2. **Ecosystem Coin Identity** – used within governance, learning, protocol, treasury, and dashboard environments to distinguish functional classifications.

Functionally, GHC is structured into two primary economic categories:

1. **Utility Coin (UC)** — governance-bearing tokens earned through platform participation
2. **Market Coin (MC)** — market-enabled tokens intended for transfer, exchange, and economic activity

The **Utility Coin (UC)** category is further segmented to support education-based mining and governance stability:

- **Mined Utility Coin (m-UC):** earned by Green Heroes through Learning and Engagement
- **Provisional Utility Coin (p-UC):** a provisional UC classification applicable to underaged participants during the Non-Execution Period
- **Virtual Utility Coin (v-UC):** a system-level governance reserve exercised by the BGH Organization

The **Market Coin (MC)** category includes:

- **Market Coin-Original (MC-o):** MC issued as part of the initial market allocation
- **Market Coin-Converted (MC-c):** formerly UC that becomes market-enabled upon unlocking

9.2 Core Token Principles

9.2.1 Utility Coin (UC)

Utility Coin (UC) exists to support sustainability participation and decentralized governance within the BGH Program.

(a) Mined Utility Coin (m-UC)

Mined Utility Coin (m-UC) refers to GHC earned by Green Heroes (GHs) through Learning and Engagement activities on the BGH Platform.

- Earning m-UC is defined as **Mining**
- Mining requires completion of educational units and successful assessment outcomes
- Upon mining, m-UC is automatically placed into a **locked** state
- Locked m-UC carries governance eligibility by default, subject to protocol rules

(b) Provisional Utility Coin (p-UC)

Provisional Utility Coin (p-UC) is a provisional classification of Utility Coin applied to underaged Green Heroes during the applicable Non-Execution Period.

- p-UC represents a provisional form of m-UC
- p-UC is non-executable and non-commercial by classification
- Governance participation associated with p-UC is defined exclusively in Section 6

(c) Unlocking and UC-to-MC Conversion

When a holder unlocks Utility Coin:

- the token exits the UC category
- it is automatically reclassified as **Market Coin-Converted (MC-c)**
- MC-c carries **no governance rights** and does not participate in Proposals or Voting

Important Notes on Unlocking

1. Intent Not Required

Unlocking may occur with or without intent to trade, exchange, or transfer GHC in the financial market.

2. **Partial Unlocking Permitted**

A holder may unlock part or all of their m-UC holdings.

3. **Voting Power Reduction**

Unlocked m-UC immediately loses Voting Power (VPo) in the equivalent quantity unlocked.

Any remaining locked m-UC retains Voting Power.

4. **Irrevocability**

Unlocking is irreversible. Unlocked GHC cannot be re-locked.

(d) Virtual Utility Coin (v-UC)

Virtual Utility Coin (v-UC) is a governance mechanism designed to protect early-stage platform stability and scientific integrity.

- v-UC is exercised by the BGH Organization
- v-UC participates in governance voting via Voting Power (VPo)
- v-UC decreases over time as community-mined m-UC increases

9.2.2 Market Coin (MC)

Market Coin (MC) is designed for economic activity and circulation.

- MC may be traded, transferred, or exchanged
- MC does not carry Voting Power (VPo)
- MC-converted (MC-c) exists when UC is unlocked and released into market-enabled form

Once unlocking occurs, the token is treated as market-enabled and non-governance-bearing, regardless of whether further market action is taken.

9.3 Total Issuance and Allocation Framework

9.3.1 Total Token Supply

Total Green Hero Coin (GHC) issuance is capped at:

9.5 billion (9.5B) GHC

This quantity is conceptually aligned with the estimated global population around the year 2046, reinforcing the Program's mission of scalable global participation.

9.3.2 50/50 Allocation Rule

Total GHC supply follows a foundational 50/50 allocation:

- **Utility Coin (UC):** 4.75B GHC
- **Market Coin (MC):** 4.75B GHC

9.3.2.1 Utility Coin Design Rationale

The UC allocation is designed to correspond to the estimated “educable” global population capable of engaging with science-based sustainability learning at scale.

9.4 Utility Coin Distribution and Mining Controls

9.4.1 Mining Controls (Mining Limitation Schedule – MLS)

To ensure long-term sustainability and equitable participation, m-UC mining is rate-limited per user.

For regular users:

- Daily: 2 (up to ~5)
- Weekly: 10 (up to ~20)
- Monthly: 30 (up to ~60)
- Yearly: 200 (up to ~400)

For subscribed users:

- Daily: 5 (up to ~10)
- Weekly: 20 (up to ~40)
- Monthly: 60 (up to ~120)
- Yearly: 400 (up to ~800)

Total mined m-UC is capped at **4.75B**.

9.4.2 Virtual Utility Coin (v-UC) Supply Logic

v-UC is defined as the remaining portion of the Utility Coin allocation not yet mined:

$$\mathbf{v\text{-}UC = 4.75B - cumulative\ mined\ m\text{-}UC}$$

This calculation does not reverse-count UC later converted into MC-c.

9.4.3 UC Redistribution Mechanism

As global participation increases and m-UC is mined, v-UC decreases in equal proportion. This mechanism enables gradual decentralization of governance authority from the BGH Organization to the community.

9.4.4 Launch Incentives (Rollout Merit Points)

At Official Public Launch (OPL), Rollout Merit Points may be applied to incentivize early participation:

- First 0 ~ 0.5M: $1 \text{ GHC} \times 3 = 1.5\text{M}$
- First 0.5M ~ 1.0M: $1 \text{ GHC} \times 2 = 1.0\text{M}$

Total Rollout Merit Points: 2.5M GHC

9.5 Market Coin Distribution and Treasury Framework

9.5.1 Founder' Allocation

At launch, the founder, Sae Chang, is entitled with 0.5B GHCs

9.5.2 Founder's Release Limitation

To support long-term stability:

- The founder may release no more than **5% of their initial allocation per year**
- Full release spans approximately **20 years from OPL**

This limitation applies only to the initial allocation.

9.5.3 Treasury Balance

Remaining MC allocation is retained by the BGH Treasury.

At OPL:

Treasury Balance = 4.75B – 0.5B – (pre-OPL allowance)

If pre-OPL allowance is 0.6B:

Treasury Balance ≈ 3.65B GHC

9.5.4 Treasury Allowance

Treasury Allowance is the portion of Treasury-held MC available for release, transfer, or exchange.

Prior to OPL, allowances may be deployed for:

- ICO-related distribution
- Promotion and ecosystem activation aligned with the BGH Program

9.5.5 Monthly Market Coin Release (MMCR)

MC is released gradually over time:

- Target horizon: ~20 years (240 months)
- Estimated monthly release:

$$\text{MMCR} \approx 3.65\text{B} \div 240 \approx 15.2\text{M GHC/month}$$

Transfers may be executed on the **1st day of each month at 12:00 AM (ET)**.

9.6 Execution Period Targets

During the The Execution Period (TEP), the BGH Program aims to achieve:

1. **4.75B m-UC** mined globally through Learning and Engagement
2. **4.75B MC** fully released and circulated through Treasury-controlled mechanisms

9.7 Inflationary and Deflationary Dynamics (Theoretical Model)

Two opposing forces influence UC and MC quantities:

- **Deflationary force:** UC decreases as m-UC is unlocked and converted into MC-c
- **Inflationary force:** MC increases through Treasury release and UC-to-MC conversion

Despite these dynamics:

- **Total GHC issuance remains capped at 9.5B**
- A substantial portion of m-UC is expected to convert into MC over time, though timing is inherently unpredictable

9.6 Estimation of MC quantity in circulation in 6 scenarios – **TO BE UPDATED**

9.6.1 Scenario 1 ~ 3

(Condition 1) 1million (1M) m-UCs mined in Year 1 with the following variances

- a. Annual increase rate (Scenario 1) = 10%
- b. Annual increase rate (Scenario 2) = 25%
- c. Annual increase rate (Scenario 3) = 50%

(Condition 2) m-UCs converted to MCs at 50% rate

(Estimation) Total MC quantity in circulation

- a. Scenario 1 = Close to 3.5B
- b. Scenario 2 = Close to 3.5B
- c. Scenario 3 = Close to 6.1B

9.6.2 Scenario 4 ~ 6

(Condition 1) 5million (5M) m-UCs mined in Year 1 with the annual increase rate in the same variances as the (Condition 1) of 9.6.1 for Scenario 4, 5 and 6 respectively

(Condition 2) m-UCs converted to MCs at 50% rate

9.6.2.3 Total MC quantity in circulation

- a. Scenario 1 = Close to 4B
- b. Scenario 2 = Close to 4B
- c. Scenario 3 = Close to 6.1B

9.6.3 In-depth analysis provided in APPENDIX - **Estimation of the MC quantity in circulation in three scenarios** for all yearly figures from Year 1 to Year 20 in the following categories

- a. Scenario 1 ~ 3 v-UC VPo vs. m-UC VPo Estimation
- b. Scenario 4 ~ 6 v-UC VPo vs. m-UC VPo Estimation
- c. Scenario 1 ~ 3 Total MC in Circulation
- d. Scenario 4 ~ 6 Total MC in Circulation
- e. Scenario 1 ~ 3 m-UC Outflow
- f. Scenario 4 ~ 6 Total m-UC Outflow

Section 10. Treasury and The Treasury Protocol (TTP)

This section defines the **BGH Treasury** and **The Treasury Protocol (TTP)**, which govern how Market Coin (MC) is held, released, and deployed to support ecosystem execution, operational continuity, and long-term sustainability of the BGH Program.

10.1 Treasury Definitions

10.1.1 Treasury

The *Treasury* is the central custody smart contract deployed on the Internet Computer Protocol (ICP). It serves as the primary on-chain mechanism for securely holding and managing digital assets designated for program execution, with particular emphasis on the management of Market Coin (MC).

10.1.2 Treasury Balance

Treasury Balance refers to the total quantity of Market Coin (MC) held within the Treasury smart contract, including uncirculated supply remaining under Treasury control.

The Treasury Balance is reduced only when MC is transferred out of the Treasury smart contract, exchanged, or otherwise distributed through an approved execution action.

10.1.3 Treasury Allowance

Treasury Allowance is the authorized, spendable portion of the Treasury Balance. It defines the quantity of MC that is available for immediate deployment, transfer, or exchange under governance-approved Treasury actions.

10.1.4 Treasury Balance vs. Treasury Allowance Logic

Treasury operations follow the logic outlined below:

(a) Allocation (Unlocking Availability)

When a scheduled Monthly Market Coin Release (MMCR) occurs, the Treasury Allowance increases programmatically. This action expands spending authorization but does not reduce the total Treasury Balance.

(b) Execution (Actual Spending)

The Treasury Balance decreases only when MC is actually transferred from the Treasury smart contract, exchanged, or distributed through an approved Treasury execution.

10.2 Treasury Operational Principles

10.2.1 Purpose of Market Coin (MC) Deployment

The primary purpose of MC deployment is to support ecosystem execution through controlled economic and operational activity, including but not limited to:

- market trading, transfer, and exchange
- hiring, staffing, and contributor incentives
- promotional, educational, and growth initiatives
- rewards, airdrops, and ecosystem engagement events
- other uses approved as beneficial to execution of the BGH Program

All MC deployment is subject to governance approval and Treasury Protocol constraints.

10.2.2 Treasury Governance and Authority

Treasury actions are governed by the BGH governance framework described in Section 6 and are executed under the authority and responsibility of the Operating Committee (OC).

All Treasury actions involving MC release, transfer, exchange, or deployment must be approved through the applicable governance voting process prior to execution.

Governance participation in Treasury decisions **does not constitute commercial activity** and does not grant individual participants direct access to Treasury assets.

10.2.3 Value Support Objective

The Treasury Protocol (TTP) is designed to support long-term ecosystem sustainability, operational continuity, and market confidence by ensuring that Treasury deployment decisions are structured, transparent, and strategically aligned with the mission of the BGH Program.

As an indirect outcome of Treasury actions, eligible Green Heroes may benefit economically when they convert Utility Coin (UC) holdings into Market Coin (MC) form and participate in market activity in accordance with protocol rules and applicable laws. Such economic participation is **separate from governance participation** and is not guaranteed.

10.3 Treasury Allocations and Release Events

10.3.1 Pre-Launch Treasury Allowance

Prior to, or at, the Official Public Launch (OPL), the Treasury designates an initial Market Coin (MC) allowance of:

0.6B MC

This allowance may be deployed through governance-approved execution for purposes including:

- Initial Coin Offering (ICO)–related distribution
- Monthly Market Coin Release (MMCR) activation
- rewards, incentive distributions, and airdrops
- operational requirements of the BGH Organization, including staffing and strategic initiatives

10.3.2 Release Schedule

The schedule governing redistribution of MC from Treasury Balance into Treasury Allowance follows the Market Coin release framework defined in Section 9 (Tokenomics).

10.3.3 Release Timing

Treasury allocation and release events are executed on a standardized schedule at:

12:00 AM Eastern Time (Canada & USA)

The first scheduled Treasury release action occurs on the day of the Official Public Launch (OPL).

Section 11. Continuous Operations Protocol (COP)

This section defines the framework by which the **BGH Organization** maintains continuous operations after the conclusion of its primary revenue-generating mechanism through **Market Coin (MC) Release**. It further explains how operational continuity is achieved while preserving decentralized governance principles through the **Safeguard Protocol (SGP)** defined in **Section 12**.

11.1 Continuous Operations Definitions

The **BGH Organization**'s primary programmed revenue stream is derived from the issuance of **Market Coins (MCs)**, capped at a maximum cumulative issuance of **3.60 billion (3.60B) MCs**, as defined in **Section [MC Issuance and Supply Cap]**.

Upon completion of this issuance, the Organization will no longer generate revenues through MC Release. Without a predefined continuity framework, this transition could pose risks to operational sustainability.

The **Continuous Operations Protocol (COP)** establishes principles and mechanisms intended to support ongoing operational costs while maintaining alignment with the mission, governance, and decentralization objectives of the BGH Program.

11.2 Operational Principles

11.2.1 Rules Modified After the Execution Period (TEP)

- a. The principle of “no external influence,” as defined in **Section [Original Independence Clause]**, shall no longer apply following the conclusion of **The Execution Period (TEP)** defined in **Section [Execution Period Definition]**.

After **TEP**, the **BGH Organization** may engage in lawful, transparent activities to fund its continued operations, subject to the constraints defined in **Section 11.2.2**.

Such activities may include, but are not limited to:

- **Strategic partnerships** with external entities, including governments, private organizations, academic institutions, NGOs, and foundations, where financial or in-kind support is provided;
- **Commercial or quasi-commercial activities** involving the lawful use, licensing, or transfer of assets, including intellectual property, educational content, software, data, real estate, or services;

- **Sponsorships, grants, donations, and program-based funding** aligned with the BGH mission;
 - **Membership-based services, subscriptions, or educational offerings** consistent with non-profit operations.
 - All such activities are conducted exclusively by the BGH Organization or its authorized counterparts and do not confer commercial participation rights to individual Green Heroes.
- b. The **Operational Committee (OC)** may estimate the anticipated depletion of MC-based revenues and plan continuity measures in advance to avoid abrupt or disruptive funding discontinuities.
- c. Decisions made by the **OC** to ensure continuity of operations , including adjustments to MC Release timing, introduction of subscription or membership fees, or other funding mechanisms , are considered ordinary operational decisions within the scope of the Organization’s non-profit mandate.
- d. All operational changes must comply with the principles set forth in **Section 11.2.2**.

11.2.2 Rules Unchanged

Notwithstanding the flexibilities introduced under the **COP**, the following principles remain unchanged throughout the lifecycle of the **BGH Program**:

a. **Science-based education model**

All educational content and initiatives remain grounded in **scientific evidence, peer-reviewed research, and transparent methodologies**, as defined in **Section [Education Framework]**.

b. **Decentralized governance model**

Decision-making authority remains decentralized and exercised by Green Heroes through the Proposal and Voting mechanisms of the BGH Platform, as defined in **Section [Governance and Voting Model]**.

Section 12. Safeguard Protocol (SGP)

This section defines the **Safeguard Protocol (SGP)**, a preprogrammed governance mechanism designed to preserve decision-making integrity and reduce manipulation risks as governance-related token supply becomes increasingly concentrated over the lifecycle of the BGH Program.

12.1 Safeguard Definitions

As the cumulative quantity of mined Utility Coins (m-UCs) approaches its maximum issuance, the direct governance influence of the BGH Organization diminishes. At advanced issuance stages, a relatively small remaining quantity of unallocated m-UCs, particularly if most of existing m-UCs are converted to MC, may create conditions under which governance outcomes become disproportionately influenced by narrow or coordinated interests.

The **Safeguard Protocol (SGP)** establishes automated, protocol-level measures intended to mitigate such risks while preserving decentralization, transparency, and long-term governance stability.

12.2 Operational Principles

12.2.1 Safeguard Activation Threshold

The Safeguard Protocol is automatically activated when the remaining unallocated m-UC supply reaches **5% or less of the total m-UC issuance**, as defined in **Section [m-UC Supply Definition]**.

Given a total m-UC issuance of **4.75 billion (4.75B)**, this threshold corresponds to:

237.5 million m-UCs (5% of 4.75B)

12.2.2 Safeguard Procedure

Upon activation of the Safeguard Protocol, the following actions are executed automatically through the BGH smart contract:

- a. Green Heroes are ranked according to the **cumulative quantity of GHCs earned** since inception. The highest-ranking Green Heroes whose combined GHC balances equal **237.5 million GHCs** are identified.
- b. Each identified Green Hero is granted **Voting Power (VPo)** equivalent to the **total quantity of Utility Coins (UCs) earned through Learning and Engagement since**

inception through Safeguard activation, regardless of whether such m-UCs are currently locked, unlocked, converted to Market Coins (MCs), or otherwise utilized.

- c. The identified Green Heroes are designated as **Safeguard Validators** of the BGH Program and retain governance participation rights on an ongoing basis, subject to applicable inactivity rules defined in **Section 13**.
- d. Eligibility for Safeguard Validator status may be recalculated dynamically for each Proposal and Voting event based on updated rankings.

12.2.3 Safeguard Validator List (SVL)

- a. A **Safeguard Validator List (SVL)** is generated **anew for each Proposal and Voting event**, based on the most recent ranking defined in **Section 12.2.2(a)**.
 - b. Where multiple Green Heroes share the same cumulative GHC score at the minimum cutoff threshold, **all such Green Heroes are included** in the SVL. As a result, the total cumulative GHCs represented by the SVL are expected to **equal or exceed 237.5 million GHCs**.
 - c. The confirmation timing for determining the SVL follows the approval-threshold and governance scheduling rules defined in **Section [Proposal Approval and Voting Thresholds]**.
-

Section 13. Governance Continuity and Inactive Participation Management (GCIPM)

This section defines protocol-level mechanisms designed to preserve effective decentralized governance participation within the BGH Program by addressing prolonged inactivity, including but not limited to permanent disengagement or death, without relying on identity verification or off-chain assumptions.

13.1 Rationale

Governance rights within the BGH Program are earned through the accumulation of Green Hero Coins (GHCs) and may be amplified under the **Safeguard Protocol (Section 12)**. Over extended time horizons, it is expected that some Green Heroes will cease active participation.

If unaddressed, prolonged inactivity may reduce effective governance participation, impair quorum efficiency, and weaken decentralized decision-making. This section establishes non-intrusive safeguards to maintain governance functionality while preserving participant autonomy.

13.2 Design Principles

The **Governance Continuity and Inactive Participation Management (GCIPM)** framework is guided by the following principles:

a. No presumption of life status or intent

The protocol does not infer or verify identity, mortality, or personal circumstances.

b. Activity-based governance evaluation

Governance participation is determined exclusively by observable, protocol-level activity.

c. Non-confiscatory treatment

No tokens, balances, or economic rights are seized, revoked, or reassigned due to inactivity.

d. Automated and decentralized enforcement

All inactivity management mechanisms are implemented through transparent, preprogrammed logic rather than discretionary intervention.

13.3 Definition of Inactivity

A Green Hero is considered inactive for governance purposes if no qualifying governance activity has been recorded for a continuous period exceeding the inactivity threshold defined in **Section 13.7**.

13.4 Inactivity Mitigation Mechanisms

13.4.1 Voting Power Dormancy

Voting Power (VPo) associated with inactive Green Heroes may be classified as **dormant** for purposes of quorum and approval-threshold calculations.

Dormant status:

- Does not affect token ownership or balances;
- Does not permanently revoke governance rights;
- Is fully reversible upon renewed qualifying activity.

13.4.2 Activity Reconfirmation

The protocol may require periodic, low-friction governance activity to maintain active status, such as voting, proposal participation, or explicit governance signaling.

Failure to perform such activity results only in dormancy, not forfeiture.

13.4.3 Safeguard Validator Inactivity

For Green Heroes designated as **Safeguard Validators** under **Section 12**:

- The same inactivity criteria apply unless modified through decentralized governance;
- Dormant Safeguard Validator Voting Power is excluded from Safeguard quorum calculations;
- Eligibility may be dynamically reassessed on a per-Proposal basis.

13.5 Reactivation

Any Green Hero whose Voting Power (VPo) has entered dormant status may restore full governance participation at any time by performing a single qualifying governance activity.

Reactivation is immediate, permissionless, and penalty-free.

13.6 Privacy and Ethical Considerations

The BGH Program does not collect, process, or infer personal data, identity, or life-status indicators. All inactivity determinations rely solely on protocol-observable governance actions.

13.7 Governance Activity Definitions

For purposes of governance continuity, quorum determination, and voting power classification, **governance activity** is defined exclusively by protocol-recorded actions.

A Green Hero shall be considered inactive for governance purposes if **no qualifying governance activity** has been recorded for a continuous period exceeding **twenty-four (24) months**.

13.7.1 Qualifying Governance Activity

Qualifying governance activity includes, but is not limited to:

- a. Casting a vote in any Proposal or Voting event;
- b. Submitting, co-sponsoring, or formally endorsing a Proposal;
- c. Explicitly delegating or reaffirming voting preferences through supported governance mechanisms as defined by protocol-level governance tools available at the time;
- d. Any other governance-signaling action explicitly designated as qualifying by protocol-level rules defined in **Section [Governance and Voting Model]**.

13.7.2 Effect of Inactivity

Green Heroes meeting the inactivity condition defined above:

- May have their Voting Power classified as dormant for quorum and approval-threshold calculations;
- Retain full ownership of all GHCs, m-UCs, MCs, and associated rights;
- May restore active governance status at any time by performing a qualifying governance activity.

Inactivity classification does not constitute forfeiture, confiscation, or permanent revocation of any rights.

13.7.3 Flexibility Clause

The inactivity duration and qualifying governance activities defined in this section may be modified through decentralized governance processes in accordance with the Proposal and Voting mechanisms specified in this White Paper.



Other Related Documents Available:

- BGH Brand Usage Guideline (Tier 2, Document code: BGH-EX-WP-0002-2026-EN)
 - Secure Operations Protocol (Tier 2, Document code: BGH-EX-WP-0003-2026-EN)
 - Risk Factors, Disclaimers and Legal Notices (Tier 2, Document code: TBD)
 - Conflict of Interest - Hiring and Personnel Proposal Purpose (Tier 2, Document code: TBD)
 - Terms of Use (Tier 2, Document code: TBD)
 - Environmental Impact & Energy Efficiency - ICP vs Proof-of-Work Networks (Tier 3, Document code: TBD)
 - Chapter I of BE GREEN HEROES book by Sae Chang - Free distribution version (Tier 3, Document code: TBD)
 - Investor Briefing 1 – Program Overview (Tier 3, Document code: TBD)
 - Investor Briefing 2 – BGH Token How It is Different (Tier 3, Document code: TBD)
 - Investor Briefing 3 – Market Impact of BGH Chapter I Free Distribution (Tier 3, Document code: TBD)
 - Elevator Pitches (Tier 3, Document code: TBD)
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APPENDIX – A: Disclaimer

Key Risks & No Guaranteed Returns

Green Hero Token is designed as a **utility token** to incentivize science-based environmental learning and participation in the Green Hero ecosystem. It is **not** intended to represent equity, a security, or any form of guaranteed investment product.

- **No guaranteed financial returns**

The value of Green Hero Tokens may fluctuate significantly and can go down as well as up. There is **no guarantee** that tokens earned or acquired will increase in value, retain value, or be tradeable in any specific market or jurisdiction.

- **Market and liquidity risk**

Token price and liquidity depend on external markets, exchanges, and community demand. There is no assurance that sufficient liquidity will exist for token holders to buy, sell, or exchange tokens at any time.

- **Regulatory and legal uncertainty**

Crypto assets are subject to evolving laws and regulations that vary by country and region. Changes in regulatory frameworks may affect the legality, usability, or exchangeability of Green Hero Tokens.

- **Execution and governance risk**

The success of the Green Hero initiative depends on the continued development of the platform, effective treasury and governance decisions, and sustained community engagement. There is no guarantee that the roadmap, protocols, or initiatives described in this document will be implemented as planned or within any specific timeline.

Nothing in this white paper should be interpreted as financial, legal, tax, or investment advice.

Prospective participants and token holders should **conduct their own research** and, where appropriate, consult independent professional advisors before making any decisions related to Green Hero Token.

APPENDIX – B: 2046 World Population Estimate & “Educable” Reach Assumptions

This Appendix documents (i) the demographic basis for the **9.5B total GHC issuance cap**, and (ii) the assumptions used to estimate the “**educable**” vs. “**currently unreachable**” share of the global population under different delivery and accessibility conditions.

B.1 2046 World Population Estimate (Data Anchor)

The BGH Program references a projected global population of approximately **9.5 billion people** in the year **2046**, consistent with widely cited projections based on the **United Nations World Population Prospects (medium variant scenario)**.

For practical clarity and consistency across this White Paper:

Projected world population in 2046 (UN medium variant): ≈ 9.5 billion people

B.2 “Educable” Population: Concept and Approach

In the BGH Program, the term “**educable population**” is used as a practical planning concept to estimate the portion of the world population that is realistically reachable for science-based education at scale, given real-world constraints such as:

- minimum age readiness
- literacy levels
- foundational STEM readiness
- delivery and access constraints
- language availability and localization
- accommodation needs for accessibility

This framework is not intended to exclude any group as inherently “**uneducable**.” Instead, it highlights implementation barriers that can be reduced over time through improved delivery, localization, scaffolding, and inclusive design.

B.3 Assumptions (With Data Anchors and Scenario Ranges)

The following assumptions are applied as multiplicative filters to avoid double-counting overlapping constraints.

B3.1 Baseline Population

- **World population baseline (2025):** approximately **8.1–8.2B**

B3.2 Age Readiness

- Content is designed for approximately **lower-secondary and above** (roughly age **12+**).
- A practical “too-young” filter of approximately **20%** is applied as a reasonable global estimate.

B3.3 Literacy (Ages 12+)

- Literacy is modeled under scenario bounds of approximately:
 - **80% (conservative) to 90% (optimistic)**

B3.4 Foundational STEM Readiness

Foundational readiness is modeled under scenario bounds of approximately:

- **65% (conservative) to 85% (optimistic)**

This factor is lower for more quantitative content (e.g., lifecycle assessment).

B3.5 Delivery and Access

If delivery relies primarily on digital systems, reach may be constrained by global internet access.

Scenario bounds applied:

- **60% (conservative) to 90% (optimistic)**

Mixed delivery strategies (schools + print + low-bandwidth options) materially improve reach.

B3.6 Language Availability

Language availability is treated as a major program lever rather than a fixed global constant.

Scenario bounds applied:

- **60% (conservative) to 90% (optimistic)**

B3.7 Accessibility and Specialized Supports

A meaningful share of the population experiences significant disabilities or learning difficulties requiring accommodations. Without inclusive design, some participants may become functionally *unreachable*.

Practical reach loss assumptions:

- **3–5% loss** in conservative/base cases
- **~2% loss** in optimistic cases (with strong accommodations)

B4 Scenario Model and Results (*Reachable* vs. *Unreachable*)

The reach model is computed multiplicatively:

Reachable Share =

(Age-ready) × (Literate) × (Foundational readiness) × (Access/Delivery) × (Language) × (Accommodation factor)

***Unreachable* Share = 1 – Reachable Share**

B4.1 Track 1 — Basic Polymer Science

Scenario	<i>Reachable</i>	<i>Unreachable</i> (“currently unreachable”)
Optimistic	~52%	~48%
Base Case	~32%	~68%
Conservative	~13.5%	~86.5%

B4.2 Track 2 — Textile LCA (Cultivation → Manufacturing → Disposal)

This track is modeled as slightly more difficult due to higher numeracy and systems-thinking requirements.

Scenario	<i>Reachable</i>	<i>Unreachable</i> (“currently unreachable”)
Optimistic	~48.6–49%	~51%
Base Case	~29.6%	~70.4%
Conservative	~12.4%	~87.6%

B5 Key Design Levers That Improve Reach

The following design actions are expected to materially reduce the “**currently unreachable**” share:

1. Localization and readability

- translation into major languages
- simplified reading levels where appropriate
- glossaries, visuals, and structured learning support

2. Access beyond broadband

- print/PDF materials
- low-bandwidth web options
- teacher guides and offline pathways

3. STEM scaffolding (especially for LCA)

- stepwise numeracy ramps
- units, ratios, uncertainty, and worked examples

4. Inclusive design

- captions, transcripts
- dyslexia-friendly formatting
- optional pacing tools and accessibility support

5. Age-flexible on-ramps

- pre-modules for ages ~10–12 using stories and applied analogies

B6 Updated Language Strategy: Translation Into Major Languages (≥50M Speakers)

The BGH Program intends to translate major learning materials (including BE GREEN HEROES and COLD WINTER WARM WINTER) into major world languages with large speaker populations.

This includes the initially listed languages:

- French, Italian, Spanish, German, Arabic, Russian, Chinese, Korean, Japanese

And expands to include additional widely spoken languages (by total speakers), including:

- English, Hindi, Bengali, Portuguese, Urdu
- Indonesian/Malay, Turkish, Vietnamese, Persian (Farsi)
- Punjabi, Marathi, Telugu, Tamil, Javanese, Thai
- Hausa, Swahili
(among others as required for global coverage)

B6.1 Recalculated “*Currently Unreachable*” Shares (Language Improved Only)

Assuming the language barrier is materially reduced via broad translation coverage, while other constraints remain unchanged, the updated “*currently unreachable*” shares are estimated as:

Basic Polymer Science

Scenario	<i>Unreachable (Updated)</i>
Optimistic	~47–48%
Base Case	~66–67%
Conservative	~80%

Textile LCA

Scenario	<i>Unreachable (Updated)</i>
Optimistic	~50–51%
Base Case	~69%
Conservative	~81%

Interpretation: Translation improves reach, but the dominant constraints remain age readiness, foundational numeracy, delivery access, and accommodation readiness.

B7 Notes on Uncertainty

These estimates are global blended values and vary significantly by country due to differences in:

- demographic age structures
- schooling completion rates
- internet and infrastructure access
- language distribution and localization capacity

Additionally, literacy and learning readiness are trend-moving variables rather than fixed constants, and may improve materially through localized partnerships and implementation strategies.

B8 Headline Estimate: Population Influencing Textile Purchases (2025)

BGH recognizes that textile sustainability education is broadly relevant because most of the world either purchases textiles directly or influences those decisions.

A practical estimate of the global population exerting active influence on textile purchases is:

~6.6 billion people (range: ~5.9B to ~6.75B)

B8.1 Transparent Calculation Logic (Summary)

This estimate combines:

- adults (18+) with significant decision autonomy
- plus a portion of older children/teens (10–17) who influence household purchasing decisions

This estimate is intended as a planning reference and not a strict demographic boundary.

Appendix B - Summary

- **2046 world population estimate: ≈ 9.5B**

- **Optimistic-case “currently unreachable” share today: ~50%** depending on track difficulty
- **Translation into major languages reduces *unreachable* share modestly**, because the largest constraints remain delivery access, STEM readiness, age readiness, and accommodations
- **Global influence on textile purchases (2025): ~6.6B** people (approx.)



APPENDIX – C: Priority Target Countries (Two-Tier Model)

This appendix defines BGH’s initial target-country framework for **Milestone 2** described in **Section 2** of this White Paper. It replaces broad phrasing such as “*countries around the world*” with a structured, updateable rollout cohort called **Priority Target Countries (PTCs)**.

C1. Definition: Priority Target Countries (PTCs)

Priority Target Countries (PTCs) are defined as national markets prioritized for early institutional adoption of BGH’s textile education because they exert outsized influence over (i) global textile consumption and/or (ii) the global expansion of sustainability narratives and knowledge related to textiles.

The PTC framework is designed to remain practical, execution-oriented, and governance-compatible. Accordingly, the PTC cohort may be refined over time through BGH’s governance mechanisms, subject to the constraints outlined in the Governance and Protocol sections of this White Paper.

C2. Objective of the Target Country Strategy (Milestone 2 Alignment)

BGH’s Milestone 2 rollout strategy is designed to maximize **global information expansion** - the rapid, credible spread of science-based textile sustainability knowledge into public discourse and real-world decision-making.

PTCs are therefore selected to prioritize countries that materially shape one or more of the following:

- **Demand-side impact**, including large-scale consumption dynamics that influence upstream production, retail systems, and lifecycle environmental outcomes; and
- **Cross-border knowledge diffusion**, including education systems, cultural networks, media reach, and institutional pathways that determine how environmental narratives spread internationally.

This approach allows BGH to focus first on countries where early curriculum adoption can generate **strong multiplier effects**, accelerating the spread of science-based sustainability frameworks beyond national boundaries.

Producer-country and waste-receiving-country rollouts remain important for long-term environmental outcomes, but they require different operational and governance strategies and may therefore be addressed in later phases under separate milestones or implementation programs.

C3. Selection Criteria (Defensible and Updateable)

BGH defines PTCs as countries that satisfy one or more of the following criteria. These criteria are intended to remain **transparent, measurable, and updateable** to reflect changes in global conditions over time.

C3.1 Information Amplification Potential (Science Diffusion Leverage)

Countries with strong ability to accelerate cross-border information expansion through one or more of the following:

- scalable education systems,
- institutional capacity for curriculum adoption,
- strong media and publishing reach,
- high digital connectivity, or
- significant linguistic, diaspora, or cultural diffusion pathways.

C3.2 Absolute Demand-Side Influence (Aggregate Market Scale)

Countries with large apparel markets and high total textile demand, shaping global supply-chain scale, production incentives, and market-driven environmental impacts.

C3.3 High-Intensity Consumption Patterns (Per-Capita Impact Pressure)

Countries with high per-capita apparel consumption and/or high discretionary spending on fashion—often correlated with faster trend cycles, higher textile turnover, and increased waste intensity.

C3.4 Narrative and Cultural Influence (“Soft Power”)

Countries that materially shape global beliefs and narratives about fashion, textiles, and sustainability through cultural export, internationally influential brands, luxury ecosystems, fashion capitals, and creative industries.

C3.5 Global Coverage Requirement (Continental Representation)

To ensure credibility as a global initiative and prevent regionally biased rollout structures, the PTC cohort includes representation across all major continents.

C4. Two-Tier PTC Structure (50 Countries Total)

BGH organizes the PTC cohort into two tiers:

- **PTC Tier 1 (Core 20):** Highest-impact consumer and influence markets
- **PTC Tier 2 (Extended 30):** Additional high-relevance markets for global coverage and scaling

This tiering helps BGH set clearer milestones (e.g., “adopted in X of Tier 1 by date Y”) while maintaining a credible global rollout framework.

C4.1 PTC Tier 1 — Core 20 (Highest Impact)

North America (2)

1. United States
2. Canada

South America (1)

3. Brazil

Europe (7)

4. United Kingdom
5. France
6. Germany
7. Italy
8. Spain
9. Netherlands
10. Switzerland

Africa (1)

11. South Africa

Asia (7)

12. China
13. India
14. Japan
15. South Korea
16. Indonesia
17. Turkey
18. United Arab Emirates

Oceania (2)

19. Australia
20. New Zealand

Rationale: Tier 1 includes major global apparel markets, high-intensity consumption markets, and high soft-power fashion hubs. It is designed to capture the largest immediate leverage points for changing global textile sustainability behavior and discourse.

C4.2 PTC Tier 2 — Extended 30 (Scale and Global Coverage)

North America (1)

21. Mexico

South America (5)

22. Argentina
23. Chile
24. Colombia
25. Peru
26. Ecuador

Europe (11)

27. Belgium
28. Sweden
29. Norway
30. Denmark

- 31. Ireland
- 32. Austria
- 33. Poland
- 34. Portugal
- 35. Greece
- 36. Czech Republic
- 37. Romania

Africa (6)

- 38. Nigeria
- 39. Egypt
- 40. Kenya
- 41. Morocco
- 42. Ghana
- 43. Ethiopia

Asia (7)

- 44. Vietnam
- 45. Thailand
- 46. Philippines
- 47. Malaysia
- 48. Singapore
- 49. Saudi Arabia
- 50. Israel

Rationale: Tier 2 expands regional coverage and scaling potential while maintaining the focus on markets with meaningful consumption levels and/or influence within their regions.

C5. Governance, Review Cycle, and Updates

The PTC list is a practical execution cohort and may be refined over time based on:

- updated market and consumption data
- measured educational outcomes and adoption success
- localization readiness and delivery constraints
- proposals and voting outcomes under BGH governance protocols

This approach ensures the target-country strategy remains mission-aligned, data-responsive, and globally scalable without locking BGH into static geographic assumptions.



APPENDIX – D: Rationale for Governance Inactivity Management

D.1 Purpose of This Appendix

This appendix provides the rationale for the governance inactivity framework defined in Section 13 (Governance Continuity and Inactive Participation Management), with particular focus on the 24-month inactivity criterion introduced in Section 13.7 (Governance Activity Definitions).

The purpose of this appendix is explanatory only. It does not introduce additional rules, obligations, or protocol behavior beyond what is defined in the core sections of this White Paper.

D.2 Governance Longevity and Real-World Constraints

The BGH Program is designed as a long-term, mission-driven initiative focused on science-based environmental education and decentralized participation. Unlike short-cycle financial protocols or high-frequency governance systems, BGH governance decisions are expected to occur over extended time horizons and may not require frequent participant interaction.

Over such long durations, it is reasonable to expect that some Green Heroes will eventually become inactive due to factors such as:

- Voluntary disengagement;
- Loss of access credentials;
- Shifts in personal priorities;
- Permanent inability to participate, including death.

Absent a defined framework, permanently inactive participants could remain indefinitely counted in governance calculations, gradually reducing effective participation rates and impairing quorum efficiency.

D.3 Why Activity-Based Criteria Are Used

The BGH Program does not attempt to determine the identity, life status, or intent of any participant. Instead, governance continuity relies exclusively on protocol-observable activity.

This design choice reflects the following principles:

- Respect for privacy and anonymity;
- Avoidance of off-chain verification or subjective judgment;
- Compatibility with decentralized and permissionless participation.

Accordingly, inactivity is defined solely by the absence of governance participation, not by any personal or off-chain condition.

D.4 Rationale for the 24-Month Inactivity Period

The 24-month inactivity threshold was selected to balance governance effectiveness with inclusiveness and fairness.

Key considerations include:

- Long-term participation tolerance: Contributors may reasonably disengage for extended periods without abandoning alignment with the mission.
- Avoidance of premature exclusion: Shorter inactivity periods could unintentionally penalize infrequent but committed participants.
- Low governance frequency: Given the nature of BGH governance, meaningful participation does not require frequent action.
- Reversibility: Inactivity results only in temporary dormancy of voting power, not forfeiture or loss of rights.

The chosen period reflects a conservative and participant-friendly approach, minimizing the risk of unjust exclusion while maintaining governance functionality.

D.5 Non-Punitive and Reversible Design

The inactivity framework is explicitly non-punitive:

- No tokens are confiscated, revoked, or reassigned;
- Ownership and economic rights remain unaffected;
- Voting Power may be reactivated immediately upon renewed participation.

This ensures that inactivity management serves governance efficiency rather than enforcement or control.

D.6 Interaction with the Safeguard Protocol

The inactivity framework operates in conjunction with the Safeguard Protocol (Section 12), which is designed to preserve governance integrity at late stages of token issuance.

By allowing inactive voting power to enter a dormant state while enabling dynamic recalculation of Safeguard Validators, the combined design:

- Prevents long-term concentration of inactive influence;
- Maintains informed and mission-aligned decision-making;
- Preserves decentralization even as participation patterns evolve.

D.7 Future Adjustability

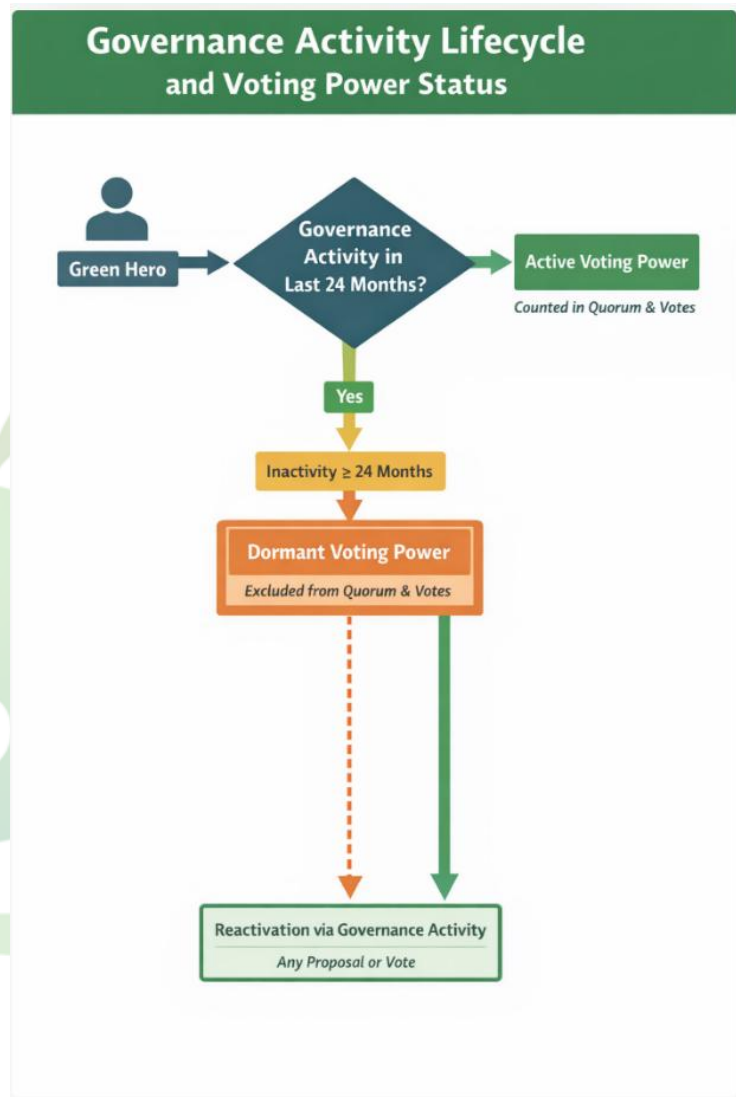
The BGH Program recognizes that participation patterns may change over time. For this reason, inactivity thresholds and qualifying governance activities may be adjusted through decentralized governance processes, as defined in the core protocol.

This ensures adaptability without requiring fundamental redesign or unilateral intervention.

D.9 Governance Activity Lifecycle (Conceptual Diagram)

Diagram title: Governance Activity Lifecycle and Voting Power Status

Diagram Structure



D.10 Step-by-Step Flow Explanation

This section explains the governance inactivity mechanism illustrated above.

Step 1 - Green Hero Participation

Each Green Hero earns governance participation rights through the accumulation of GHCs, as defined in Section [Governance and Voting Model]. Voting Power (VPo) is associated with these rights and may be amplified under the Safeguard Protocol (Section 12).

Step 2 - Governance Activity Monitoring

The BGH Protocol continuously monitors qualifying governance activity, defined in Section 13.7, including:

- Voting on Proposals;
- Submitting or endorsing Proposals;
- Explicit governance signaling actions.

No off-chain identity checks or personal data are used.

Step 3 - Active Status (Default)

If at least one qualifying governance activity is recorded within the preceding 24-month period, the Green Hero's Voting Power remains active and fully counted toward:

- Quorum calculations;
- Approval thresholds;
- Safeguard-related governance processes.

Step 4 - Inactivity Threshold Reached

If no qualifying governance activity is recorded for a continuous period exceeding 24 months, the Green Hero is classified as inactive for governance purposes.

This classification:

- Does not imply forfeiture;
- Does not affect token ownership;
- Does not restrict future participation.

Step 5 - Voting Power Dormancy

Upon inactivity classification:

- Voting Power enters a dormant state;
- Dormant Voting Power is excluded from quorum and approval-threshold calculations;
- All tokens and associated rights remain fully intact.

This step prevents long-term accumulation of inactive voting weight from impairing governance effectiveness.

Step 6 - Special Case: Safeguard Validators

For Green Heroes designated as Safeguard Validators under Section 12:

- The same inactivity criteria apply unless modified by governance-approved updates;
- Dormant Safeguard Voting Power is excluded from Safeguard quorum calculations;
- Validator eligibility may be dynamically reassessed per Proposal.

This ensures that elevated governance roles remain aligned with active participation.

Step 7 — Reactivation

At any time, an inactive Green Hero may restore full active Voting Power by performing a single qualifying governance activity.

Reactivation:

- Is immediate;
- Requires no approval;
- Does not impose penalties or reductions;
- Restores full participation rights.

D.11 Design Outcomes Illustrated by the Flow

This flow ensures that:

- Governance remains effective even over long program lifecycles;
- Inactive participation does not silently degrade quorum quality;
- No assumptions are made about personal circumstances;
- All governance rights remain recoverable and non-custodial.

APPENDIX – E: Scenarios for Annual m-UC & MC Quantity Estimations

Conditions applicable for all three scenarios

1. 1M MUCs mined in Year 1
2. UC converted to MC at 50% rate
3. ICO quantity of 0.1B
4. Monthly MC Release quantity is 15.2M
5. Founders and Treasury Initial Balance of 1B released in the first 10 year period

Scenario - 1: Annual MUC increase rate is **10%**

Year	(A) Annual MUC Quantity at compound rate of 10%	(B) MUC Annual Cumulative	MC Quantity in Circulation				(T=C+D+E+F) Total MC Quantity (Cumulative)
			(C) MUC converted to MC at 50% rate (Cumulative)	(B) MUC converted + (D) ICO (100M)	(E) MMC Released and Accumulation (15.2M/month , Cumulative)	(F) Founders + Treasury Initial Balance (1B released in 10 years)	
1	1,000,000	1,000,000	500,000	100,500,000	182,400,000	100,000,000	382,900,000
2	1,100,000	2,100,000	1,050,000	101,050,000	364,800,000	100,000,000	565,850,000
3	1,210,000	3,310,000	1,655,000	101,655,000	547,200,000	100,000,000	748,855,000
4	1,331,000	4,641,000	2,320,500	102,320,500	729,600,000	100,000,000	931,920,500
5	1,464,100	6,105,100	3,052,550	103,052,550	912,000,000	100,000,000	1,115,052,550
6	1,610,510	7,715,610	3,857,805	103,857,805	1,094,400,000	100,000,000	1,298,257,805
7	1,771,561	9,487,171	4,743,586	104,743,586	1,276,800,000	100,000,000	1,481,543,586
8	1,948,717	11,435,888	5,717,944	105,717,944	1,459,200,000	100,000,000	1,664,917,944
9	2,143,589	13,579,477	6,789,738	106,789,738	1,641,600,000	100,000,000	1,848,389,738
10	2,357,948	15,937,425	7,968,712	107,968,712	1,824,000,000	100,000,000	2,031,968,712
11	2,593,742	18,531,167	9,265,584	109,265,584	2,006,400,000		2,115,665,584
12	2,853,117	21,384,284	10,692,142	110,692,142	2,188,800,000		2,299,492,142
13	3,138,428	24,522,712	12,261,356	112,261,356	2,371,200,000		2,483,461,356
14	3,452,271	27,974,983	13,987,492	113,987,492	2,553,600,000		2,667,587,492
15	3,797,498	31,772,482	15,886,241	115,886,241	2,736,000,000		2,851,886,241
16	4,177,248	35,949,730	17,974,865	117,974,865	2,918,400,000		3,036,374,865
17	4,594,973	40,544,703	20,272,351	120,272,351	3,100,800,000		3,221,072,351
18	5,054,470	45,599,173	22,799,587	122,799,587	3,283,200,000		3,405,999,587
19	5,559,917	51,159,090	25,579,545	125,579,545	3,465,600,000		3,591,179,545
20	6,115,909	57,274,999	28,637,500	128,637,500	3,648,000,000		3,776,637,500

Scenario - 2: Annual MUC increase rate is 25%

Year	(A) Annual MUC Quantity at compound rate of 25%	(B) MUC Annual Cumulative	MC Quantity in Circulation				(T=C+D+E+F) Total MC Quantity (Cumulative)
			(C) MUC converted to MC at 50% rate (Cumulative)	(B) MUC converted + (D) ICO (100M)	(E) FMMC Released and Accumulation (12.5M/month, Cumulative)	(F) Founders + Treasury Initial Balance (1B released in 10 years)	
1	1,000,000	1,000,000	500,000	100,500,000	182,400,000	100,000,000	382,900,000
2	1,250,000	2,250,000	1,125,000	101,125,000	364,800,000	100,000,000	565,925,000
3	1,562,500	2,812,500	1,406,250	101,406,250	547,200,000	100,000,000	748,606,250
4	1,953,125	3,515,625	1,757,813	101,757,813	729,600,000	100,000,000	931,357,813
5	2,441,406	4,394,531	2,197,266	102,197,266	912,000,000	100,000,000	1,114,197,266
6	3,051,758	5,493,164	2,746,582	102,746,582	1,094,400,000	100,000,000	1,297,146,582
7	3,814,697	6,866,455	3,433,228	103,433,228	1,276,800,000	100,000,000	1,480,233,228
8	4,768,372	8,583,069	4,291,534	104,291,534	1,459,200,000	100,000,000	1,663,491,534
9	5,960,464	10,728,836	5,364,418	105,364,418	1,641,600,000	100,000,000	1,846,964,418
10	7,450,581	13,411,045	6,705,523	106,705,523	1,824,000,000	100,000,000	2,030,705,523
11	9,313,226	16,763,806	8,381,903	108,381,903	2,006,400,000		2,114,781,903
12	11,641,532	20,954,758	10,477,379	110,477,379	2,188,800,000		2,299,277,379
13	14,551,915	26,193,447	13,096,724	113,096,724	2,371,200,000		2,484,296,724
14	18,189,894	32,741,809	16,370,905	116,370,905	2,553,600,000		2,669,970,905
15	22,737,368	40,927,262	20,463,631	120,463,631	2,736,000,000		2,856,463,631
16	28,421,709	51,159,077	25,579,538	125,579,538	2,918,400,000		3,043,979,538
17	35,527,137	63,948,846	31,974,423	131,974,423	3,100,800,000		3,232,774,423
18	44,408,921	79,936,058	39,968,029	139,968,029	3,283,200,000		3,423,168,029
19	55,511,151	99,920,072	49,960,036	149,960,036	3,465,600,000		3,615,560,036
20	69,388,939	124,900,090	62,450,045	162,450,045	3,648,000,000		3,810,450,045

Scenario - 3: Annual MUC increase rate is 50%

Year	(A) Annual MUC Quantity at compound rate of 50%	(B) MUC Annual Cumulative	MC Quantity in Circulation				(T=C+D+E+F) Total MC Quantity (Cumulative)
			(C) MUC converted to MC at 50% rate (Cumulative)	(B) MUC converted + (D) ICO (100M)	(E) FMMC Released and Accumulation (12.5M/month, Cumulative)	(F) Founders + Treasury Initial Balance (1B released in 10 years)	
1	1,000,000	1,000,000	500,000	100,500,000	182,400,000	100,000,000	382,900,000
2	1,500,000	2,500,000	1,250,000	101,250,000	364,800,000	100,000,000	566,050,000
3	2,250,000	4,750,000	2,375,000	102,375,000	547,200,000	100,000,000	749,575,000
4	3,375,000	8,125,000	4,062,500	104,062,500	729,600,000	100,000,000	933,662,500
5	5,062,500	13,187,500	6,593,750	106,593,750	912,000,000	100,000,000	1,118,593,750
6	7,593,750	20,781,250	10,390,625	110,390,625	1,094,400,000	100,000,000	1,304,790,625
7	11,390,625	32,171,875	16,085,938	116,085,938	1,276,800,000	100,000,000	1,492,885,938
8	17,085,938	49,257,813	24,628,906	124,628,906	1,459,200,000	100,000,000	1,683,828,906
9	25,628,906	74,886,719	37,443,359	137,443,359	1,641,600,000	100,000,000	1,879,043,359
10	38,443,359	113,330,078	56,665,039	156,665,039	1,824,000,000	100,000,000	2,080,665,039
11	57,665,039	170,995,117	85,497,559	185,497,559	2,006,400,000		2,191,897,559
12	86,497,559	257,492,676	128,746,338	228,746,338	2,188,800,000		2,417,546,338
13	129,746,338	387,239,014	193,619,507	293,619,507	2,371,200,000		2,664,819,507
14	194,619,507	581,858,521	290,929,260	390,929,260	2,553,600,000		2,944,529,260
15	291,929,260	873,787,781	436,893,890	536,893,890	2,736,000,000		3,272,893,890
16	437,893,890	1,311,681,671	655,840,836	755,840,836	2,918,400,000		3,674,240,836
17	656,840,836	1,968,522,507	984,261,253	1,084,261,253	3,100,800,000		4,185,061,253
18	985,261,253	2,953,783,760	1,476,891,880	1,576,891,880	3,283,200,000		4,860,091,880
19	1,477,891,880	4,431,675,640	2,215,837,820	2,315,837,820	3,465,600,000		5,781,437,820
20	318,324,360	4,750,000,000	2,375,000,000	2,475,000,000	3,648,000,000		6,123,000,000

APPENDIX – F: Scenarios for Voting Power(VPo) Redistribution

Conditions applicable for all three scenarios

1. 1M MUCs mined in Year 1
2. UC converted to MC at 50% rate
3. ICO quantity of 0.1B
4. Monthly MC Release quantity is 15.2M
5. Founders and Treasury Initial Balance of 1B released in the first 10 year period

Scenario - 1

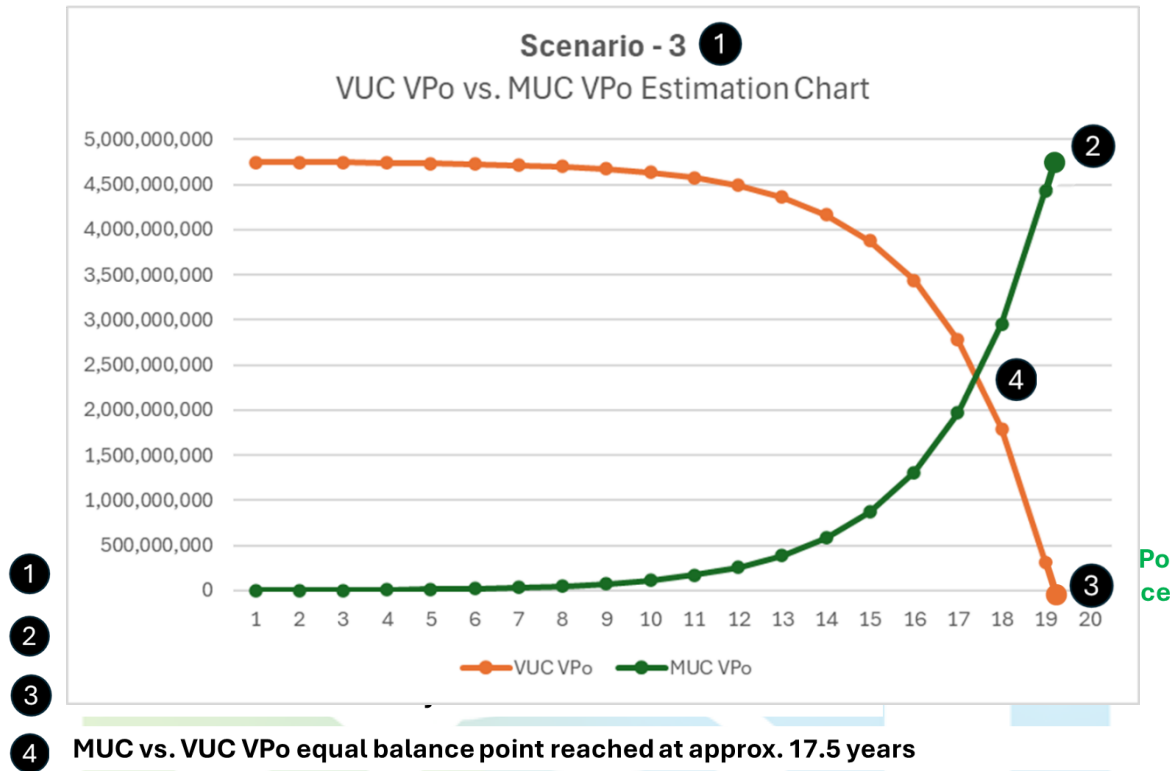
Year	MUC / VUC					Favourable VPo Regression Estimation (=(1) - (2))
	(A) Annual MUC Quantity at compound rate of 10%	(B) MUC Annual Cumulative	MUC VPo	VUC VPo	(1) Favourable VPo from MUC & VUC	
1	1,000,000	1,000,000	1,000,000	4,749,000,000	4,750,000,000	4,624,750,000
2	1,100,000	2,100,000	2,100,000	4,747,900,000	4,750,000,000	4,549,475,000
3	1,210,000	3,310,000	3,310,000	4,746,690,000	4,750,000,000	4,474,172,500
4	1,331,000	4,641,000	4,641,000	4,745,359,000	4,750,000,000	4,398,839,750
5	1,464,100	6,105,100	6,105,100	4,743,894,900	4,750,000,000	4,323,473,725
6	1,610,510	7,715,610	7,715,610	4,742,284,390	4,750,000,000	4,248,071,098
7	1,771,561	9,487,171	9,487,171	4,740,512,829	4,750,000,000	4,172,628,207
8	1,948,717	11,435,888	11,435,888	4,738,564,112	4,750,000,000	4,097,141,028
9	2,143,589	13,579,477	13,579,477	4,736,420,523	4,750,000,000	4,021,605,131
10	2,357,948	15,937,425	15,937,425	4,734,062,575	4,750,000,000	3,946,015,644
11	2,593,742	18,531,167	18,531,167	4,731,468,833	4,750,000,000	3,870,367,208
12	2,853,117	21,384,284	21,384,284	4,728,615,716	4,750,000,000	3,794,653,929
13	3,138,428	24,522,712	24,522,712	4,725,477,288	4,750,000,000	3,718,869,322
14	3,452,271	27,974,983	27,974,983	4,722,025,017	4,750,000,000	3,643,006,254
15	3,797,498	31,772,482	31,772,482	4,718,227,518	4,750,000,000	3,567,056,880
16	4,177,248	35,949,730	35,949,730	4,714,050,270	4,750,000,000	3,491,012,568
17	4,594,973	40,544,703	40,544,703	4,709,455,297	4,750,000,000	3,414,863,824
18	5,054,470	45,599,173	45,599,173	4,704,400,827	4,750,000,000	3,338,600,207
19	5,559,917	51,159,090	51,159,090	4,698,840,910	4,750,000,000	3,262,210,227
20	6,115,909	57,274,999	57,274,999	4,692,725,001	4,750,000,000	3,185,681,250

Scenario - 2

Year	MUC					Favourable VPo Regression Estimation (=(1) - (2))
	(A) Annual MUC Quantity at compound rate of 25%	(B) MUC Annual Cumulative	MUC VPo	VUC VPo	Favourable VPo from MUC & VUC	
1	1,000,000	1,000,000	1,000,000	4,749,000,000	4,750,000,000	4,624,500,000
2	1,250,000	2,250,000	2,250,000	4,747,750,000	4,750,000,000	4,548,875,000
3	1,562,500	2,812,500	2,812,500	4,747,187,500	4,750,000,000	4,473,593,750
4	1,953,125	3,515,625	3,515,625	4,746,484,375	4,750,000,000	4,398,242,188
5	2,441,406	4,394,531	4,394,531	4,745,605,469	4,750,000,000	4,322,802,734
6	3,051,758	5,493,164	5,493,164	4,744,506,836	4,750,000,000	4,247,253,418
7	3,814,697	6,866,455	6,866,455	4,743,133,545	4,750,000,000	4,171,566,772
8	4,768,372	8,583,069	8,583,069	4,741,416,931	4,750,000,000	4,095,708,466
9	5,960,464	10,728,836	10,728,836	4,739,271,164	4,750,000,000	4,019,635,582
10	7,450,581	13,411,045	13,411,045	4,736,588,955	4,750,000,000	3,943,294,477
11	9,313,226	16,763,806	16,763,806	4,733,236,194	4,750,000,000	3,866,618,097
12	11,641,532	20,954,758	20,954,758	4,729,045,242	4,750,000,000	3,789,522,621
13	14,551,915	26,193,447	26,193,447	4,723,806,553	4,750,000,000	3,711,903,276
14	18,189,894	32,741,809	32,741,809	4,717,258,191	4,750,000,000	3,633,629,095
15	22,737,368	40,927,262	40,927,262	4,709,072,738	4,750,000,000	3,554,536,369
16	28,421,709	51,159,077	51,159,077	4,698,840,923	4,750,000,000	3,474,420,462
17	35,527,137	63,948,846	63,948,846	4,686,051,154	4,750,000,000	3,393,025,577
18	44,408,921	79,936,058	79,936,058	4,670,063,942	4,750,000,000	3,310,031,971
19	55,511,151	99,920,072	99,920,072	4,650,079,928	4,750,000,000	3,225,039,964
20	69,388,939	124,900,090	124,900,090	4,625,099,910	4,750,000,000	3,137,549,955

Scenario - 3

Year	MUC					Favourable VPo Regression Estimation (=(1) - (2))
	(A) Annual MUC Quantity at compound rate of 50%	(B) MUC Annual Cumulative	MUC VPo	VUC VPo	Favourable VPo from MUC & VUC	
1	1,000,000	1,000,000	1,000,000	4,749,000,000	4,750,000,000	4,624,500,000
2	1,500,000	2,500,000	2,500,000	4,747,500,000	4,750,000,000	4,548,750,000
3	2,250,000	4,750,000	4,750,000	4,745,250,000	4,750,000,000	4,472,625,000
4	3,375,000	8,125,000	8,125,000	4,741,875,000	4,750,000,000	4,395,937,500
5	5,062,500	13,187,500	13,187,500	4,736,812,500	4,750,000,000	4,318,406,250
6	7,593,750	20,781,250	20,781,250	4,729,218,750	4,750,000,000	4,239,609,375
7	11,390,625	32,171,875	32,171,875	4,717,828,125	4,750,000,000	4,158,914,063
8	17,085,938	49,257,813	49,257,813	4,700,742,188	4,750,000,000	4,075,371,094
9	25,628,906	74,886,719	74,886,719	4,675,113,281	4,750,000,000	3,987,556,641
10	38,443,359	113,330,078	113,330,078	4,636,669,922	4,750,000,000	3,893,334,961
11	57,665,039	170,995,117	170,995,117	4,579,004,883	4,750,000,000	3,789,502,441
12	86,497,559	257,492,676	257,492,676	4,492,507,324	4,750,000,000	3,671,253,662
13	129,746,338	387,239,014	387,239,014	4,362,760,986	4,750,000,000	3,531,380,493
14	194,619,507	581,858,521	581,858,521	4,168,141,479	4,750,000,000	3,359,070,740
15	291,929,260	873,787,781	873,787,781	3,876,212,219	4,750,000,000	3,138,106,110
16	437,893,890	1,311,681,671	1,311,681,671	3,438,318,329	4,750,000,000	2,844,159,164
17	656,840,836	1,968,522,507	1,968,522,507	2,781,477,493	4,750,000,000	2,440,738,747
18	985,261,253	2,953,783,760	2,953,783,760	1,796,216,240	4,750,000,000	1,873,108,120
19	1,477,891,880	4,431,675,640	4,431,675,640	318,324,360	4,750,000,000	1,059,162,180
20	318,324,360	4,750,000,000	4,750,000,000	0	4,750,000,000	984,162,180



Glossary

A

AG

Assessment & Grading.

Protocol-level mechanisms used to verify learning outcomes, assessment results, and eligibility for earning Utility Coins through Learning and Engagement.

Author

The individual or entity holding the copyright to content proposed for inclusion in the BGH Platform.

B

BE GREEN HEROES (BGH)

The name of the global Learning-to-Earn program, ecosystem, platform, and mission described in this White Paper, focused on science-based environmental sustainability education.

BGH Organization

The non-profit entity responsible for executing, managing, and stewarding the BGH Program and operating the BGH Platform.

BGH Platform

The combined online, on-chain, and offline systems through which Learning and Engagement, governance, token operations, and program execution occur.

BGH Platform-online / BGH Platform-onchain

The digital and blockchain-based components of the BGH Platform.

BGH Platform-offline

The real-world and non-digital components of the BGH Platform, including physical events, printed materials, and institutional partnerships.

Board

The governing body of the BGH Organization formed in accordance with applicable non-profit laws, responsible for organizational oversight.

C

COP (Continuous Operations Protocol)

The protocol governing how the BGH Organization maintains operational continuity after the conclusion of Market Coin-based revenue generation.

Crypto Assets / Digital Assets

Blockchain-based representations of value, including tokens, cryptocurrencies, and other cryptographic instruments.

E

Ecosystem Coin Identity

The governance and functional representation layer used to visually distinguish Utility Coin (UC), Market Coin (MC), and their sub-classifications within the BGH ecosystem.

ET (Eastern Time)

The standard time reference used throughout this White Paper (Canada & USA).

G

GCIPM (Governance Continuity and Inactive Participation Management)

Protocol-level mechanisms designed to preserve effective decentralized governance by addressing prolonged participant inactivity without identity verification or confiscation.

GHC (Green Hero Coin)

The native token of the BGH ecosystem, used to reward verified learning, enable governance participation, and support controlled market activity.

GH (Green Hero)

A participant in the BGH Program who engages in science-based sustainability learning and related activities.

Governance

All decision-making processes related to managing the BGH Program, including Proposals, Voting, and protocol evolution.

I

ICP (Internet Computer Protocol)

The blockchain infrastructure selected for BGH's on-chain deployment due to its energy-efficient architecture.

ICO (Initial Coin Offering)

A limited early distribution mechanism for Market Coins, subject to treasury and governance constraints.

ISMC (International Sustainability Marketing Competition)

A structured engagement and learning competition operated by the BGH Organization.

L**Learning and Engagement**

The educational, participatory, and mission-aligned activities through which Green Heroes acquire knowledge and earn Utility Coins.

M**MC (Market Coin)**

A category of Green Hero Coin designed for trading, transfer, and market activity, with no governance rights.

MC-o (Market Coin-Original)

Market Coins issued directly by the BGH Organization as part of the market allocation.

MC-c (Market Coin-Converted)

Utility Coins that become Market Coins after being unlocked.

MC Release (MCR)

The act of releasing Market Coins into circulation under treasury and governance control.

m-UC (Mined Utility Coin)

Utility Coins earned by Green Heroes through verified Learning and Engagement activities.

Mining

The process of earning Utility Coins by completing educational content and assessments.

MLS (Mining Limitation Schedule)

Rate limits governing how many Utility Coins may be mined per participant over defined time periods.

MMCR (Monthly Market Coin Release)

A scheduled mechanism by which Market Coins move from Treasury Balance to Treasury Allowance.

N

NEP (Non-Execution Period)

The period during which Provisional Utility Coins remain restricted from execution, transfer, conversion, or commercial activity.

O

OC (Operating Committee)

The executive body responsible for operational execution of the BGH Program under governance oversight.

OPL (Official Public Launch)

The public commencement of the BGH Program and activation of the BGH Platform.

P

p-UC (Provisional Utility Coin)

A provisional classification of Utility Coin earned by an Underaged Green Hero, granting governance participation while prohibiting commercial execution during the Non-Execution Period.

Proposal

A formal governance submission requesting a change, addition, or adjustment to eligible aspects of the BGH Program.

Pro-A (Proposal-Approved)

A Proposal that has passed the Voting Process and is approved for implementation.

Pro-R (Proposal-Rejected)

A Proposal that has failed the Voting Process.

Proposer

The individual or entity submitting a Proposal.

S

Sae

The minimum divisible unit of Green Hero Coin, equal to 0.00000000001 GHC.

SGP (Safeguard Protocol)

An automated governance mechanism activated at late issuance stages to preserve decision-making integrity and prevent governance manipulation.

Smart Contract

A blockchain-based program used to automate governance, token mechanics, treasury actions, and protocol rules.

Support

The act of endorsing a Proposal to qualify it for Voting.

SVL (Safeguard Validator List)

The dynamically generated list of Green Heroes granted enhanced governance roles under the Safeguard Protocol.

T

TEP (The Execution Period)

The primary operational phase during which Utility Coins are mined and Market Coins are released.

TGP (The Grace Period)

A stabilization period following Official Public Launch and prior to full smart-contract immutability.

Token-Optimized Identity

The standardized circular token representation of GHC used in blockchain infrastructure, exchanges, wallets, and market-facing environments.

Treasury

The on-chain custody mechanism holding and managing Market Coins and other program assets.

Treasury Allowance

The authorized, spendable portion of the Treasury Balance.

Treasury Balance

The total quantity of Market Coins held by the Treasury.

TTP (The Treasury Protocol)

The protocol governing how the Treasury operates, releases funds, and executes approved actions.

U

UC (Utility Coin)

The governance-bearing category of Green Hero Coin earned through Learning and Engagement.

Unlocking

The irreversible action by which Utility Coins are converted into Market Coins.

u-GH (Underaged Green Hero)

A Green Hero who has not reached the minimum legal age for independent commercial participation.

V

v-UC (Virtual Utility Coin)

A non-transferable, system-level governance instrument exercised by the BGH Organization. v-UC participates in governance calculations under the Ecosystem Coin Identity layer and is structurally designed to regress proportionally as m-UC supply increases.

VPo (Voting Power)

The weighted unit used to calculate governance influence in Proposal and Voting processes.

VPt (Voting Protocol)

The rules governing how votes are cast, counted, and validated.

VPr (Voting Process)

The procedural execution of voting events within the BGH Platform.

W

White Paper

This document and all associated appendices, protocols, and referenced materials describing the BGH Program.

Wallet

A cryptographic tool used to store keys and interact with blockchain assets, including Green Hero Coin.